Materials Chemistry

Topic 15

Key ideas

Unit 49 Natural and synthetic polymers

- Structures and properties of cellulose and chitin
- Silicates **Extension**
- Thermoplastics and thermosetting plastics
- Some more addition polymers: polytetrafluoroethene, polymethyl methacrylate and poly(methyl cyanoacrylate)
- Condensation polymers: polyesters, nylons, Kevlar and urea-methanal
- A polymeric biomaterial — polylactide

Unit 50 Relationship between the structures and properties of polymers

- Explaining the properties of polymers in terms of their structures
- Plastic fabrication processes
- Problems associated with the recycling of plastic waste

Unit 51 Metals, alloys and other synthetic materials in modern life

- Metallic crystal structures
- Properties and uses of some common alloys
- Structures and applications of liquid crystals, ceramics and nanomaterials
- Choosing the right material for specific needs in daily lives

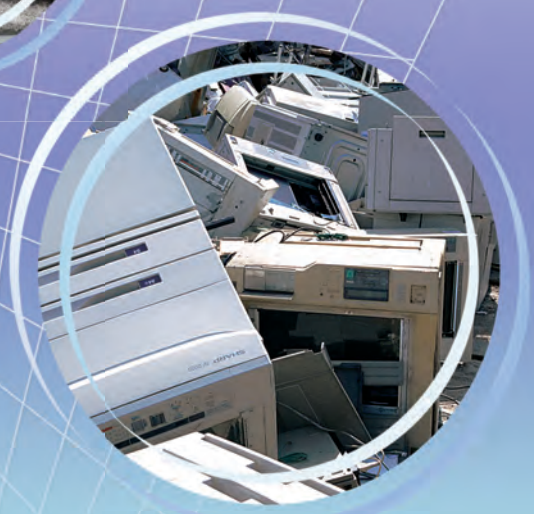
Unit 52 Green chemistry

- How green chemistry contributes to the sustainable development
- Understanding green chemistry practices









Unit 49 Natural and synthetic polymers

Pre-study
A pre-study worksheet on this unit is available for download from *Jing Kung Chemistry Website*.

49.1 Materials in life

Virtually all aspects of our daily lives — transportation, housing, clothing, communication, recreation, and food production — are influenced by materials.

Historically, the development and advancement of societies have been tied to people's ability to produce and make use of materials. Historians even divide particular periods of human history according to the metal mainly used, such as the *Copper Age*, the *Bronze Age* and the *Iron Age**.

* We have discussed this in Topic 3 Metals.

Traditionally, materials are divided into four classes: metals, **polymers**, **ceramics** and **composites**.



Fig. 49.1 Aluminium (metal) cans



Fig. 49.2 Glass (ceramic) bottles



Fig. 49.3 Plastic (polymer) bottles

A polymer is a compound consisting of very large molecules formed by joining together many small molecules repeatedly. You are probably familiar with polymers in the form of plastics*.

* We have discussed plastics in Topic 7 Fossil Fuels and Carbon Compounds.

Ceramics* are inorganic compounds of metals and non-metals. They are often oxides, nitrides or carbides. Glass, porcelain and cement are in this class.

* We will discuss ceramics in unit 51.

Copper Age 銅器時代　Bronze Age 青銅器時代　Iron Age 鐵器時代　polymer 聚合物
ceramic 陶瓷　composite 複合材料



Fig. 49.4 A catamaran's mast made of a polymer reinforced with carbon fibres

Composites are made from mixtures of the materials in the first three classes. For example, glass fibre reinforced polymers are composites. Fig. 49.5 summarizes the traditional classification of materials.

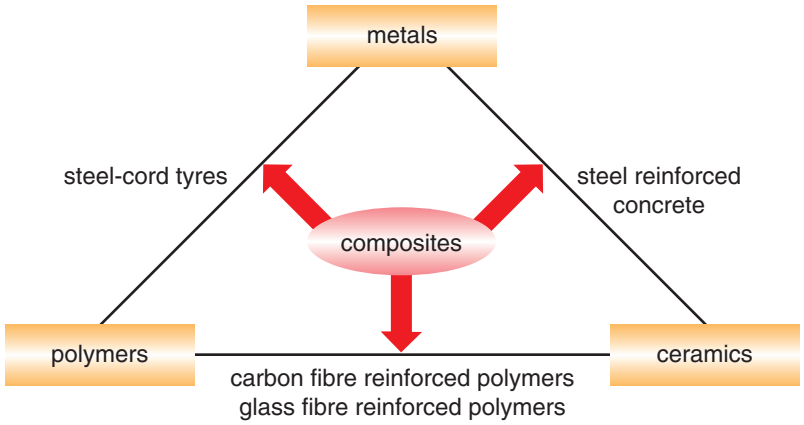
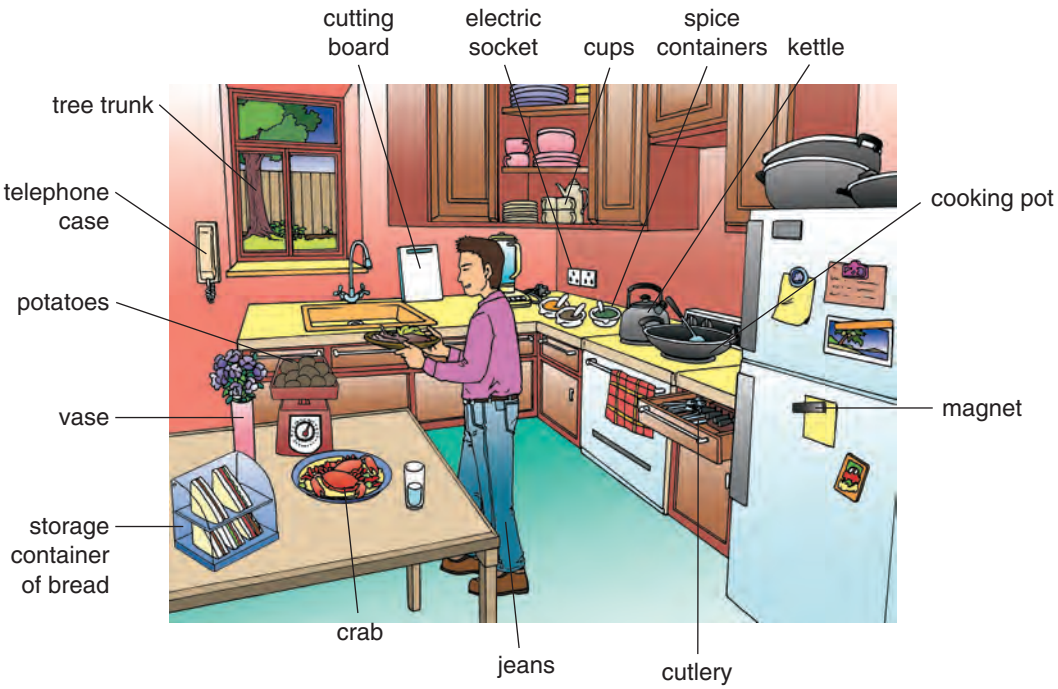


Fig. 49.5 Traditional classification of materials

Worksheet
A worksheet on this activity is available for download from *Jing Kung Chemistry Website*.

Discussion

Look at the kitchen shown below. There are electric appliances, kitchen utensils, etc.



1 Consider all the items labelled in the diagram. Cross out items that would disappear if there were

a) no metals;
Cross out cooking pot, kettle, magnet and cutlery.

b) no ceramics; **Extension**
Cross out cups, spice containers and vase.

c) no polymers.
Cross out electric socket, cutting board, tree trunk, telephone case, potatoes, storage container of bread, crab and jeans.

You may use pens of different colours for items made of different materials.

2 Refer to the items that are made of or contain polymers. Divide them into two groups:

a) those that are made of or contain natural polymers; and
Tree trunk, potatoes, crab and jeans.

b) those that are made of or contain synthetic polymers.
Electric socket, cutting board, telephone case, storage container of bread.

49.2 Polymers

We first discuss polymers in Topic 7 Fossil Fuels and Carbon Compounds. There are natural polymers and man-made polymers. Fig. 49.6 shows a classification of polymers.

* We have discussed carbohydrates and proteins in Topic 8 Chemistry of Carbon Compounds.

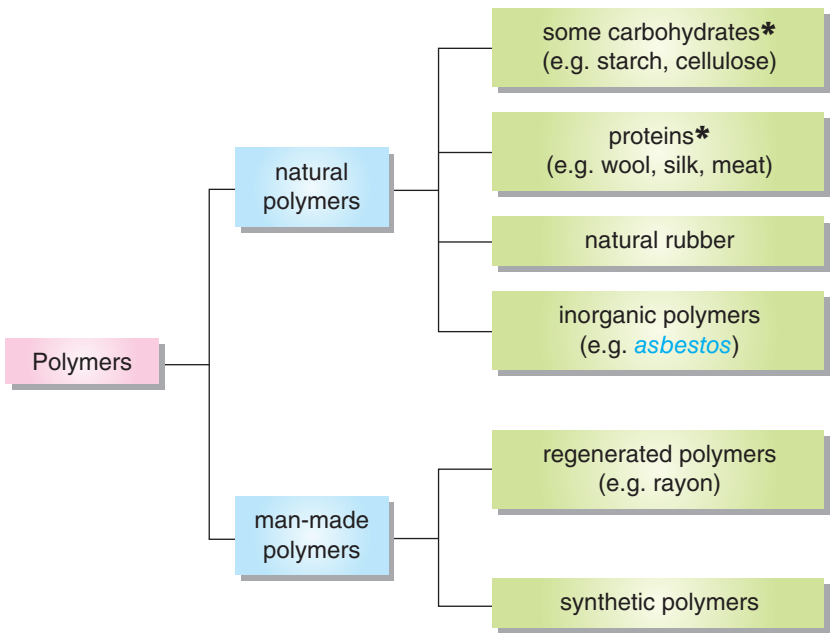


Fig. 49.6 Classification of polymers

asbestos 石棉

In Topic 7, we have discussed synthetic polymers obtained by addition polymerization. Two other types of synthetic polymers, nylons and polyesters, are learnt in Topic 8 Chemistry of Carbon Compounds. Here we are going to discuss some natural polymers and some more synthetic polymers.

49.3 Cellulose

Natural polymers such as **protein**, **starch**, **cellulose** and **chitin** play crucial roles in the lives of plants and animals.

Cellulose is the most abundant natural polymer found in the world. Cellulose is found in the cell walls of nearly all plants. It is one of the main components providing the strength and shape of plants. Cellulose also has its delicate side. **Cotton** is almost pure cellulose.



(a) (b)

Fig. 49.7 (a) Cellulose is found in the cell walls of nearly all plants
(b) Cotton is almost pure cellulose

The structure of cellulose

Fig. 49.8 shows the structure of a β-glucose molecule. The –OH group N1 at the first carbon atom (C1) is on the same side of the ring as the $-CH_2OH$ group at the fifth carbon atom (C5).

Cellulose is a natural polymer of the β-glucose monomer. Two β-glucose monomer molecules can link together via an oxygen atom at the first carbon atom (C1) of one unit and the fourth carbon atom (C4) of the next (Fig. 49.9). A water molecule is released. This is a condensation reaction. The link between the two glucose units is called a **glycosidic linkage**.

N1 Refer to page T18.

DSE 2013 Paper 2 Q2(a)(i)(2)

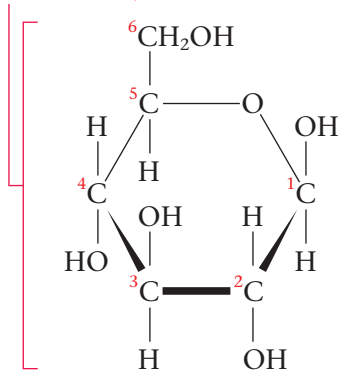


Fig. 49.8 The structure of a β-glucose molecule

protein 蛋白質 starch 澱粉 cellulose 纖維素 chitin 甲殼素
cotton 棉花 glycosidic linkage 糖苷鍵合

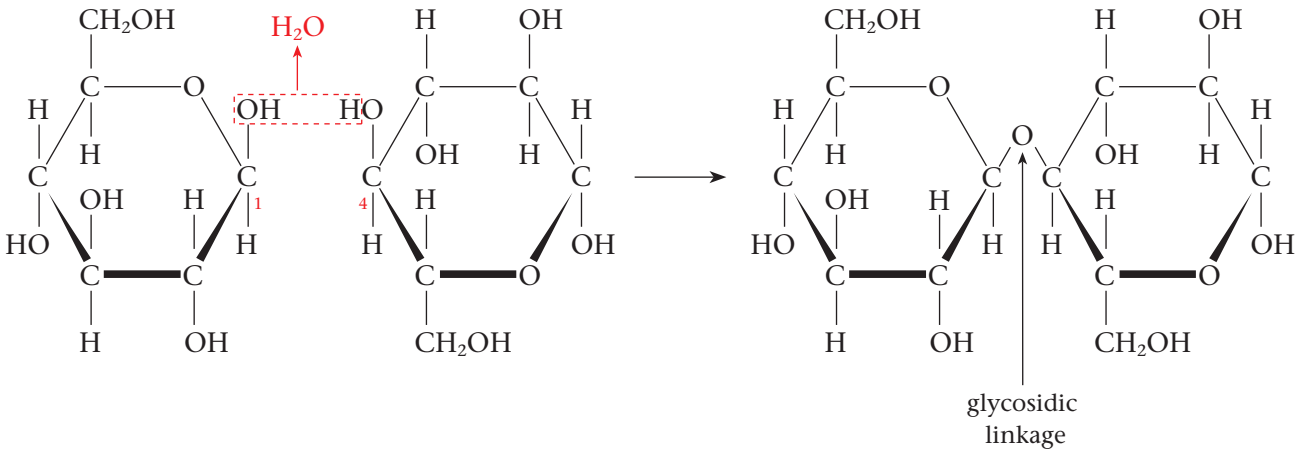


Fig. 49.9 Two β-glucose monomer molecules linked together with the release of one water molecule

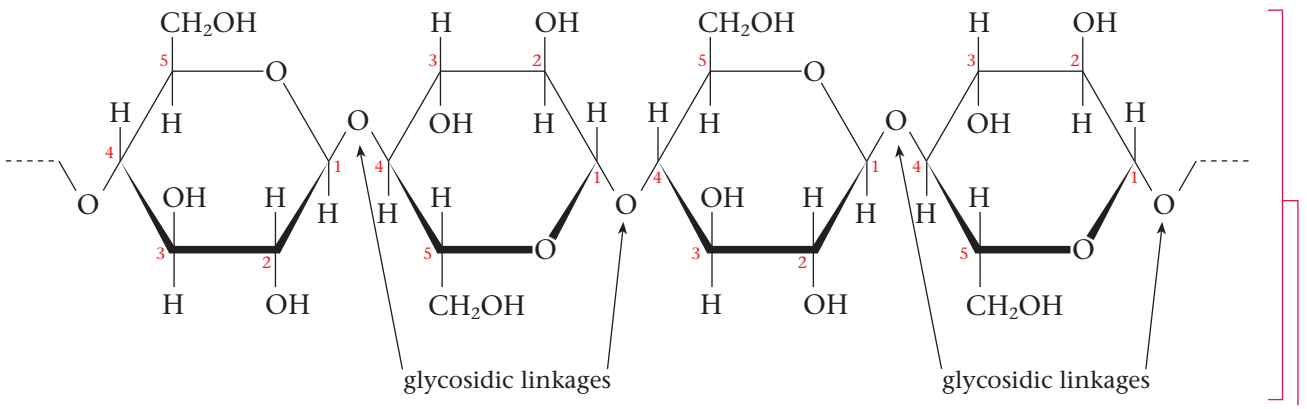
* The process is called condensation polymerization. We will discuss this later in this unit.

The dimers then continue to condense* and build up to a length of up to 10 000 glucose units arranged in a long, unbranched chain. N2

Fig. 49.10 shows the structure of a section of a cellulose chain. Notice that the $-CH_2OH$ groups on the fifth carbon atoms are placed alternately on opposite sides of the chain. This maintains a linear structure in the polymer.

N2 Range of relative molecular mass of cellulose

The general formula of cellulose is $(C_6H_{10}O_5)_n$. The relative molecular mass of cellulose generally ranges from 2.5×10^5 to 1.0×10^6. The relative molecular mass of cellulose falls into a wide range because molecules of cellulose may contain various number of glucose units.



AL 1999 Paper 2 Q6(b)(i)(II)

Fig. 49.10 The structure of a section of a cellulose chain (carbon atoms of the rings have been omitted for the sake of clarity) N3

N3 Refer to page T18.

Fig. 49.11 shows the structure of one form of cellulose. The structure consists of sheets of chains. The chains lie side by side and are held together by hydrogen bonds. The sheets are laid on top of one another in a way that staggers the chains, just as bricks are staggered to give strength and stability to a wall. Hence cellulose has a high strength.

In this structure, fewer –OH groups can interact with water molecules in the environment. As a result, cellulose is insoluble in water.

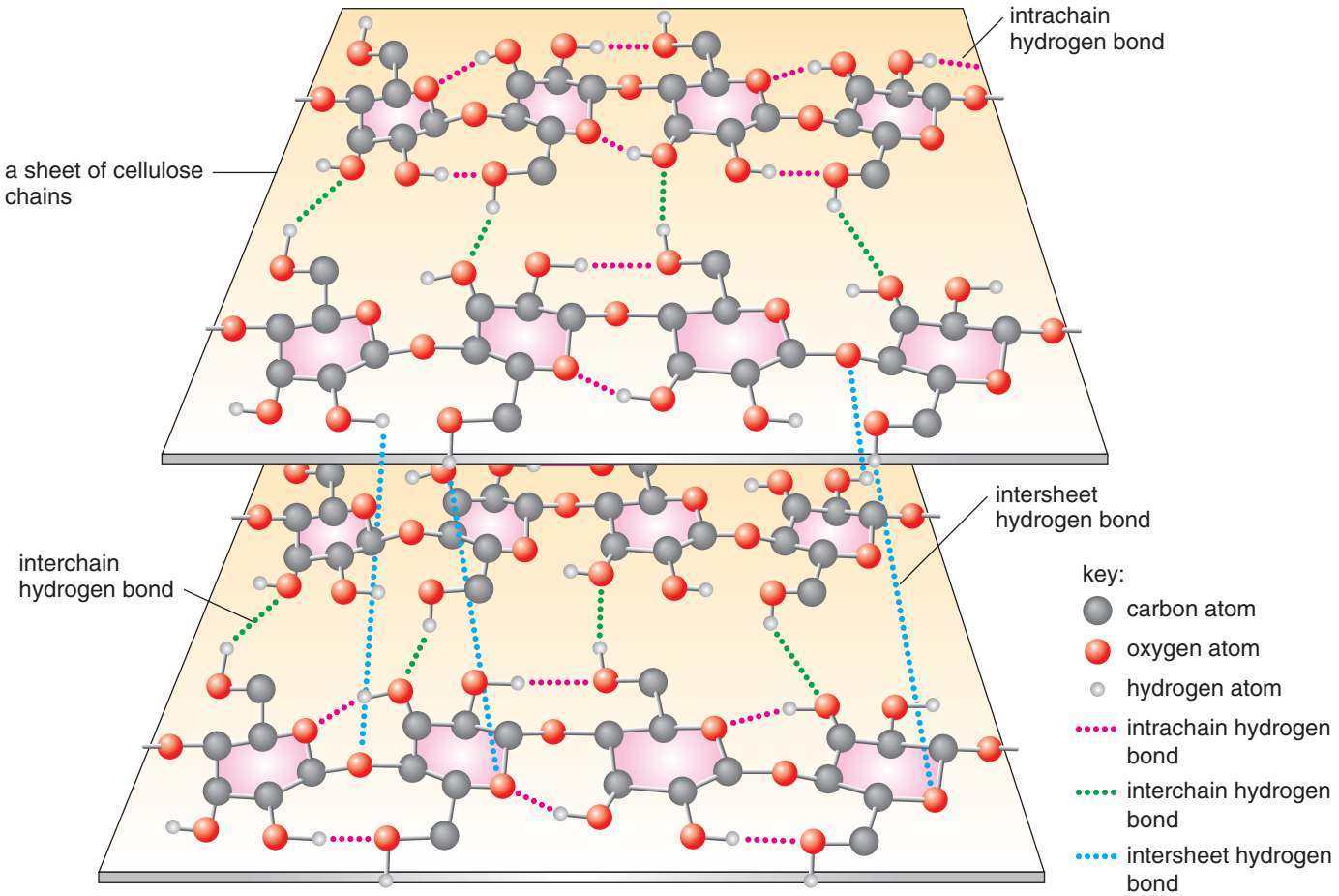


Fig. 49.11 The structure of one form of cellulose (hydrogen atoms bonded to carbon atoms are not shown)

N4 Refer to page T19.

N6 Refer to page T19.

A biodegradable polymer — cellulose N4 N5

Cellulose is a biodegradable polymer. The glycosidic linkages can be N6 broken down by microorganisms via hydrolysis.

The main component of paper is cellulose. We may use paper to make shopping bags so as to cut down the use of non-biodegradable plastics.

N5 Comparing the water solubility of glucose and cellulose

Glucose is very soluble in water while most of the celluloses are insoluble in water.

A glucose molecules has 5 –OH groups that can form hydrogen bonds with water molecules.

Many –OH groups of a cellulose molecule form hydrogen bonds with other cellulose molecules. They are less available to form hydrogen bonds with water molecules.

Checkpoint
Polyvinyl chloride can be disposed of by incineration.
Although acidic gases (such as hydrogen chloride and chlorine) would be produced, they can be removed by installing scrubbers.
Landfilling is not appropriate as polyvinyl chloride is non-biodegradable.
Cellulose can be disposed of by landfilling.
Cellulose is biodegradable. It can be decomposed by microorganisms.

Checkpoint

Large amounts of polyvinyl chloride and cellulose are used everyday. Suggest, with explanation, an appropriate waste disposal method for each material.



Fig. 49.12 Chitin is the main component of the external skeletons of spiders

49.4 Chitin

AL 1999 Paper 2 Q6(b)(ii)

A polymer that is similar to cellulose, both in its biological function and structure, is chitin. Chitin is the main component of the shells of lobsters and crabs, and the *external skeletons* of insects and spiders. Chitin is the Earth's second most abundant carbohydrate polymer (after cellulose).

Chitin is the polymer of the N-acetylglucosamine monomer (Fig. 49.13). The monomer molecules link together to form chitin in a way similar to those in cellulose. The units are also joined by glycosidic linkages.

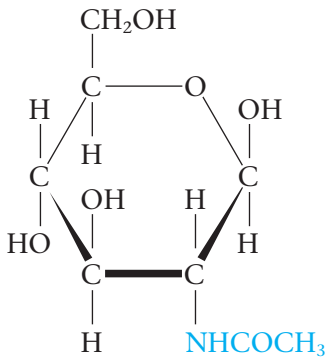


Fig. 49.13 The structure of a N-acetylglucosamine molecule

The structure of chitin is identical to that of cellulose, except that one –OH group in each glucose unit is replaced by an acetylamino group (–$NHCOCH_3$).

Like cellulose, the chains of chitin pack side by side and are held together by hydrogen bonds. Hence chitin is hard and insoluble in water. It is also biodegradable.

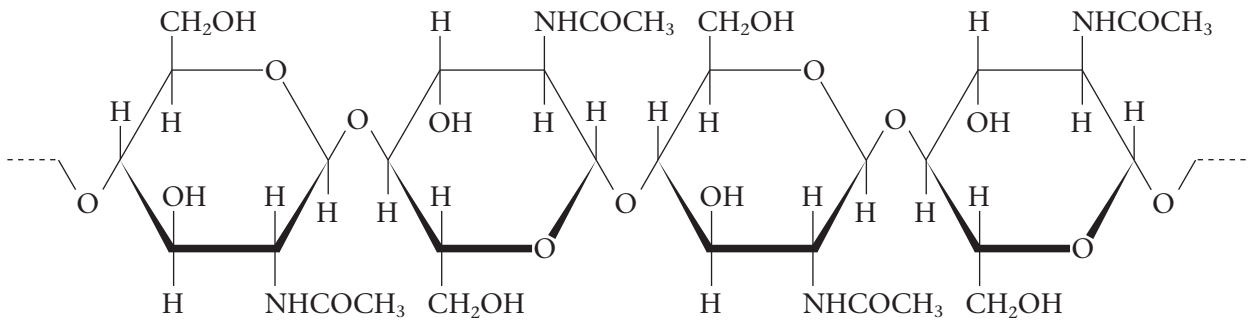


Fig. 49.14 The structure of a section of a chitin chain (carbon atoms of the rings have been omitted for the sake of clarity)

external skeleton 外骨骼

Extension

49.5 Silicates

* Rocks can be subdivided into three principal types: *igneous rocks*, *sedimentary rocks* and *metamorphic rocks*. Granite is a type of igneous rock.

The Earth's crust is mainly made up of rocks. **Granite** is a common type of rock found in Hong Kong*. Its major components include **quartz**, **feldspar** and **mica**. Quartz, feldspar and mica all belong to a group of minerals called the **silicates**, all of which contain silicon and oxygen. Silicates are minerals that comprise most (> 90%) of the Earth's crust.

Worksheet **Assessment Form**

A worksheet and an assessment form on this activity are available for download from *Jing Kung Chemistry Website*.



Internet Search & Presentation

Uses of silicon, quartz, feldspar and mica

Task Refer to page T20.

Your teacher will divide the class into groups. Each group should design a poster to present the uses of one of the following materials: silicon, quartz, feldspar and mica.

Reference websites

1 You may start your search from the following websites as appropriate:

- Uses of silicon — from Los Alamos National Laboratory's Chemistry Division
  http://periodic.lanl.gov/14.shtml
- Uses of quartz — from Mineral Information Institute
  http://www.mineraleducationcoalition.org/minerals/quartz
- Uses of feldspar — from Industrial Minerals Association of North America
  http://www.ima-na.org/?page=what_is_feldspar
- Uses of mica — from Industrial Minerals Association of North America
  http://www.mineralszone.com/minerals/mica.html

2 Extend your search using search engines and appropriate keywords.

- http://www.zoo.com/Zoo-Site/
- http://www.google.com
- http://www.yahoo.com
- http://www.searchalta.com/

granite 花崗岩　quartz 石英　feldspar 長石　mica 雲母　silicate 硅酸鹽
igneous rock 火成岩　sedimentary rock 沉積岩　metamorphic rock 變質岩

Extension

49.6 The building block of silicate minerals

All silicate minerals are built up from the basic unit of orthosilicate anion, SiO_4^{4-} (Fig. 49.15). In the anion, the silicon atom is covalently bonded to 4 oxygen atoms in the form of a tetrahedron. Each oxygen atom possesses a formal negative charge. Hence each tetrahedral unit has a formal charge of –4.

* We can represent a silicate tetrahedron by ▲ when looking from above. Remember that the corners represent oxygen atoms.

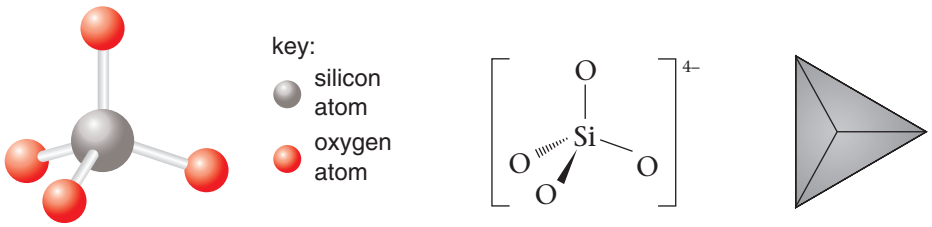


Fig. 49.15 The orthosilicate anion, SiO_4^{4-}

* Notice that the tetrahedra may join to each other by corners only, never by edges or faces.

In silicates, the SiO_4^{4-} tetrahedra are linked by sharing oxygen atoms between adjacent tetrahedra. Joining two tetrahedra* together gives the $Si_2O_7^{6-}$ ion (Fig. 49.16). Notice that each unshared oxygen carries a single negative charge.

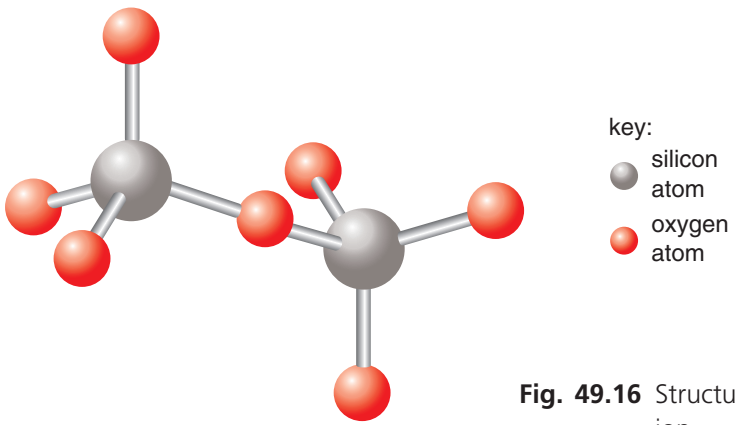
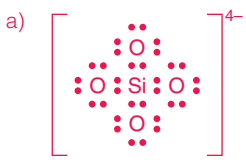


Fig. 49.16 Structure of the $Si_2O_7^{6-}$ ion

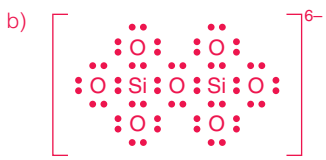
Checkpoint

Draw electron diagrams for

a) an isolated SiO_4^{4-} ion; and



b) an $Si_2O_7^{6-}$ ion made by linking two silicate tetrahedra together.



Extension

* The Si–O bond has a bond enthalpy of 466 kJ mol^{-1} while the C–C bond has a bond enthalpy of 347 kJ mol^{-1}.

The Si–O bond is particularly strong compared with the C–C bond*. The linkage –Si–O–Si–O– is very stable and instead of existing as discrete units of SiO_4^{4-} ions, silicates tend to form chains, sheets or three-dimensional networks.

When the tetrahedra are linked together, the extended units are also negatively charged. The presence of metallic ions such as Ca^{2+} or Mg^{2+} are necessary for electrical neutrality.

49.7 Chain silicates

Single chain silicates

The tetrahedra can join up to form chains by sharing two corner oxygen atoms (Fig. 49.17).



Fig. 49.17 A single chain of tetrahedra, all pointing in one direction

* Each bridging oxygen atom common to both silicon atoms counts as $\frac{1}{2}$.

In each silicate tetrahedron, two oxygen atoms are wholly owned by the silicon and the other two are shared. The number of oxygen atoms per silicon is

$$2 + \frac{1}{2} + \frac{1}{2}^{*} = 3$$

* Silicon has an oxidation state of +4 and oxygen has an oxidation state of –2.

The net charge on each SiO_3 unit is –2*. The overall structure of the chain can be represented as $(SiO_3^{2-})_n$. The repeating unit is SiO_3^{2-}.

Double chain silicates

The silicate tetrahedra can also form double chains in which two or three oxygen atoms of each tetrahedron are shared (Fig. 49.18).

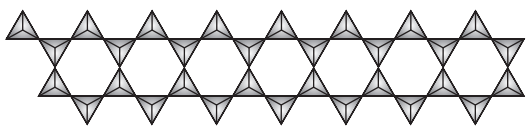


Fig. 49.18 A double chain of tetrahedra, with all their bases in a common plane and all pointing up

Extension

In each chain, half of tetrahedra are joined by three corners, half by two. Hence for half of the silicon atoms, the number of oxygen atoms per silicon is

$$1 + \frac{1}{2} + \frac{1}{2} + \frac{1}{2} = \frac{5}{2}$$

The number of oxygen atoms per silicon is 3 for the other half of silicon atoms. Thus, every two silicon atoms have $5\frac{1}{2}$ oxygen atoms and the Si:O ratio is 4:11. The overall structure can be represented as $(Si_4O_{11}^{6-})_n$. The repeating unit is $Si_4O_{11}^{6-}$.

The structure of asbestos

Asbestos is a generic term for a group of naturally occurring hydrated silicates that can be processed mechanically into long fibres. *Amphibole*, $Ca_2Mg_5(Si_4O_{11})_2(OH)_2$, is one type of asbestos. It has the double chain structure.

The electrostatic attraction between the cations and the negative silicate chains holds the structure together. Crystal *cleavage* can occur along the chain direction because the non-directional ionic links are weaker than the strong Si–O covalent bonds. Hence the asbestos has a fibrous structure (Fig. 49.19).

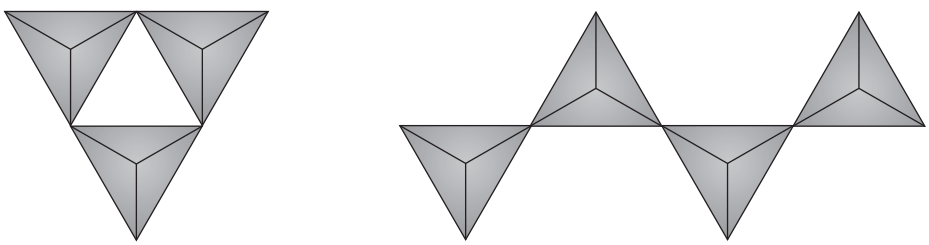


Fig. 49.19 Asbestos has a fibrous structure

Checkpoint

Two silicate ions are shown below. What is the formula and charge of each ion?

Refer to page T22.



amphibole 閃石　　cleavage 裂開

Extension

49.8 Sheet silicates

Sheets of silicate tetrahedra can be built up by joining more double chains together (Fig. 49.20). These occur in **talc** and mica.

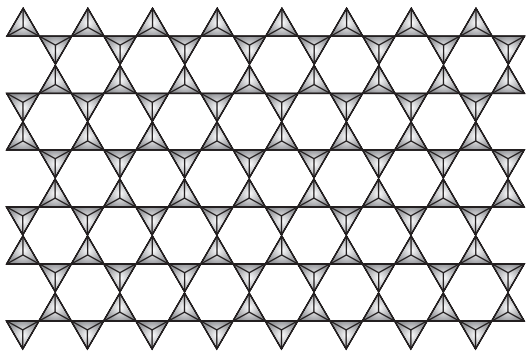


Fig. 49.20 Part of a continuous sheet of tetrahedra, with all their bases in a common plane and all pointing up

In sheets like this, all the unshared negatively charged oxygens point in one direction (upwards, out of the plane of the paper in Fig. 49.20). In each silicate tetrahedron, one oxygen atom is wholly owned by the silicon, and the other three are shared. The number of oxygen atoms per silicon is

$$1 + \frac{1}{2} + \frac{1}{2} + \frac{1}{2} = \frac{5}{2}$$

In whole numbers, the Si:O ratio is 2:5. The net charge on each Si_2O_5 unit is –2. The overall structure of the sheet can be represented as $(Si_2O_5^{2-})_n$. The repeating unit is $Si_2O_5^{2-}$.

The structure of talc

Talc, $Mg_3(Si_4O_{10})(OH)_2$, has a structure consisting of layers. Each layer is made up of parallel sheets. Talc contains two types of sheet:

- one type based on silicate tetrahedra (the tetrahedral sheet), and
- another type based on octahedra (the octahedral sheet).

Each octahedron bases on a magnesium ion surrounded by a total of six oxygen atoms and hydroxide ions in an octahedral arrangement (Fig. 49.21). The octahedra link together in a sheet (Fig. 49.22).

talc 滑石

Extension

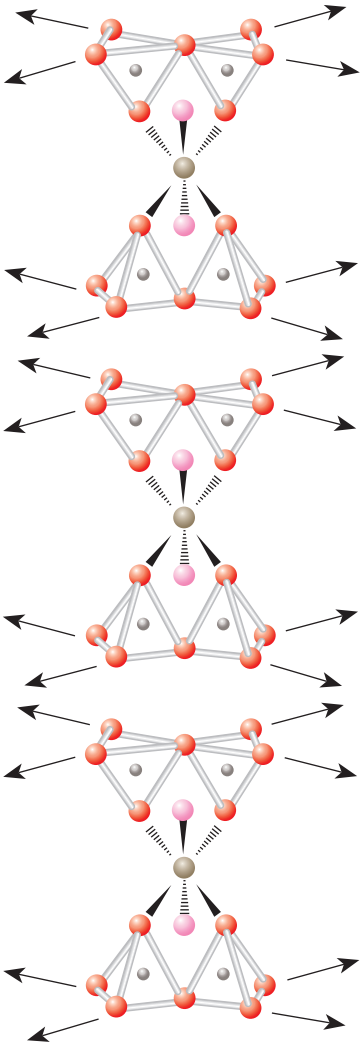


Fig. 49.23 The layers in the structure of talc (the tetrahedra are sections of silicate sheets and continue in the direction of the arrows)

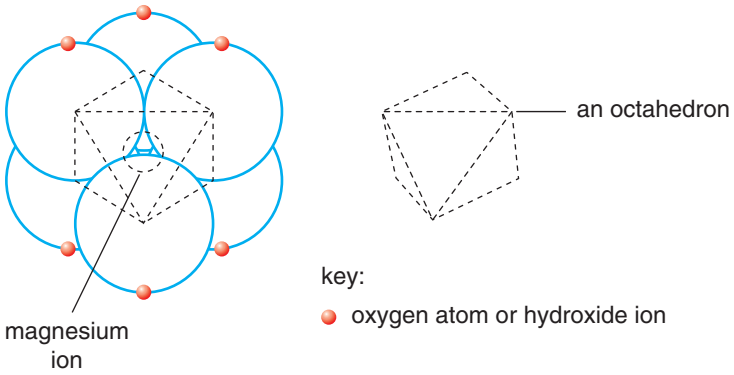


Fig. 49.21 A magnesium ion surrounded by a total of six oxygen atoms and hydroxide ions in an octahedral arrangement

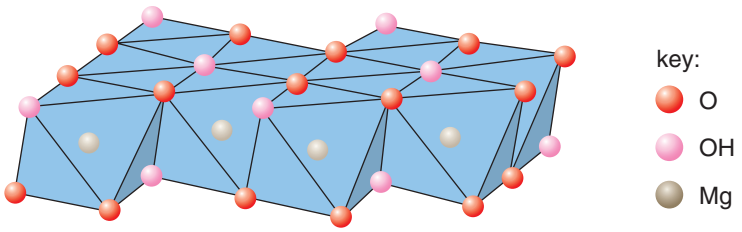


Fig. 49.22 An octahedral sheet

Fig. 49.23 shows the layers in the structure of talc. An octahedral sheet is sandwiched between the apices of two tetrahedral sheets. The apical oxygen atoms from each tetrahedral sheet also form part of one side of the octahedral sheet.

The layers in talc are not charged and are held together by weak van der Waals' forces. Thus, the layers can slide past one another easily. This explains why we can easily grind talc to make talcum powder, a soft and fine powder to make one's skin feel smooth and dry.

49.9 Network silicates

The mineral quartz is one form of **silica**. Pure quartz is a continuous three-dimensional network of silicon and oxygen atoms. In quartz, all the four vertices of each tetrahedron are linked to other tetrahedra*. Quartz is very slowly weathered to sand. As both have a giant network of strong covalent bonds, the melting points of quartz and sand are very high.

* We have discussed the structure of quartz in Topic 2 Microscopic World I.

silica 硅石

Extension

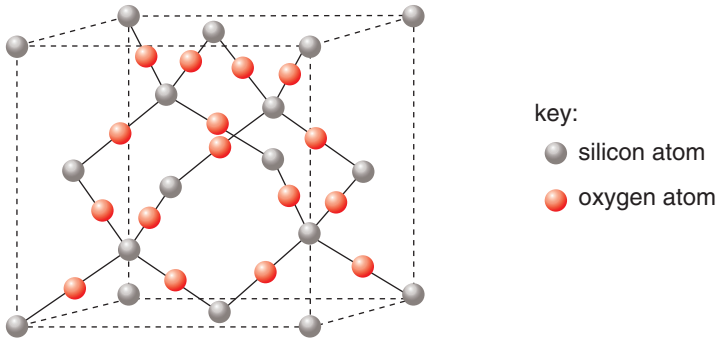


Fig. 49.24 The structure of quartz



Fig. 49.25 Crystals of quartz (SiO_2)



Fig. 49.26 Sand is made up of small grains of silicon dioxide

In quartz, all the oxygen atoms are shared. The quartz network carries no charge.

The number of oxygen atoms per silicon is:

$$\frac{1}{2} + \frac{1}{2} + \frac{1}{2} + \frac{1}{2} = 2$$

and the overall structure of quartz can be represented as $(SiO_2)_n$.

Table 49.1 summarizes some important information about chain, sheet and network silicates.

Extension

Table 49.1 Some information about chain, sheet and network silicates

Nature of structure	Single chain	Double chain	Sheet	Network
Number of corners shared for each tetrahedron	2	2.5	3	4
Representation for the overall structure	$(SiO_3^{2-})_n$	$(Si_4O_{11}^{6-})_n$	$(Si_2O_5^{2-})_n$	$(SiO_2)_n$
Example	*pyroxenes*, $CaMg(Si_2O_6)$	amphibole, $Ca_2Mg_5(Si_4O_{11})_2(OH)_2$	talc, $Mg_3(Si_4O_{10})(OH)_2$	quartz, $(SiO_2)_n$

The following activity would help you understand the structures of silicates.

Activity 49.1

Building models of silicates.

Example

49.1 Explaining the properties of talcum powder in terms of its structure

The major ingredient of baby powder is talcum powder.



Both sand and talcum powder contain silicate tetrahedra linked together. After a bath you would not sprinkle sand over your body, even if it were finely powdered. Explain why the properties of the two substances are so different.

Solution

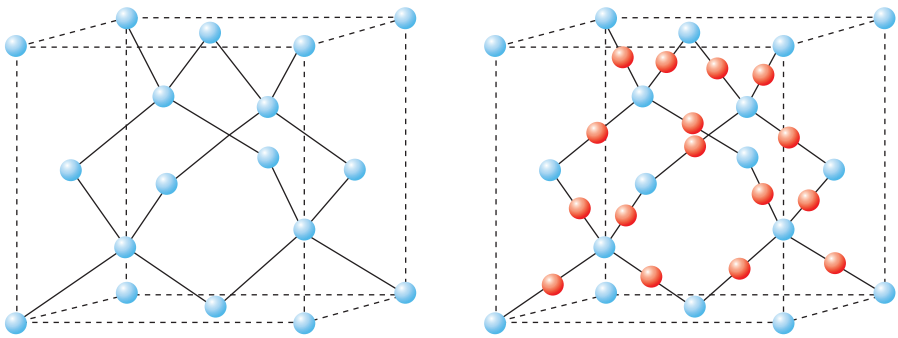
The weathered quartz crystals in sand have a three-dimensional network structure. Hence they are hard and rough. In talcum powder, the small crystals have a flat layer structure. The layers can slide past one another easily as the forces between them are weak. Thus, talcum powder is soft and has a slippery feel.

pyroxene 輝石

Extension

49.2 Studying the structures of an element and one of its oxides

X is an element in Period 3 of the periodic table. The structures of X and one of its oxides are shown below.



ASL 2004 Q6(b)

a) Suggest, with explanation, what element X is.

b) With reference to the structure of the oxide of X, explain why the oxide can be used as an *abrasive*.

ASL 2004 Q6(c)

Solution

a) In the structure of X, each atom is surrounded by four other atoms in the form of a tetrahedron, giving a giant structure similar to that of diamond. Hence X belongs to the same group as carbon. X is silicon.

b) Due to the strong covalent bonds in the oxide of X, the atoms cannot move from one place to another. Thus, the oxide is hard and strong.



Checkpoint

Checkpoint

2 Asbestos has a double chain structure. The electrostatic attraction between the cations and the negative silicate chains holds the structure together.
Crystal cleavage can occur along the chain direction because the non-direction ionic links are weaker than the strong Si–O covalent bonds. Hence asbestos has a fibrous structure.
Quartz has a giant network of strong Si–O covalent bonds. The atoms cannot move from one place to another. Thus, quartz is hard.

1 Predict the structure of each of the following silicate minerals (pairs of tetrahedra, chain, sheet, or network).

a) $Na_2ZrSi_4O_{10}$ Sheet

b) $Ca_2ZnSi_2O_7$ Pairs of tetrahedra

c) $Al_2Si_2O_5(OH)_4$ Sheets (kaolinite)

2 Asbestos and quartz are two common silicate minerals. Asbestos is a fibrous material whereas quartz occurs as hard crystals. Account for this difference in the appearance of asbestos and quartz with reference to their structures.

abrasive 磨蝕劑

Extension

* The aluminosilicate sheet can be represented as $(AlSi_3O_{10}^{5-})_n$.

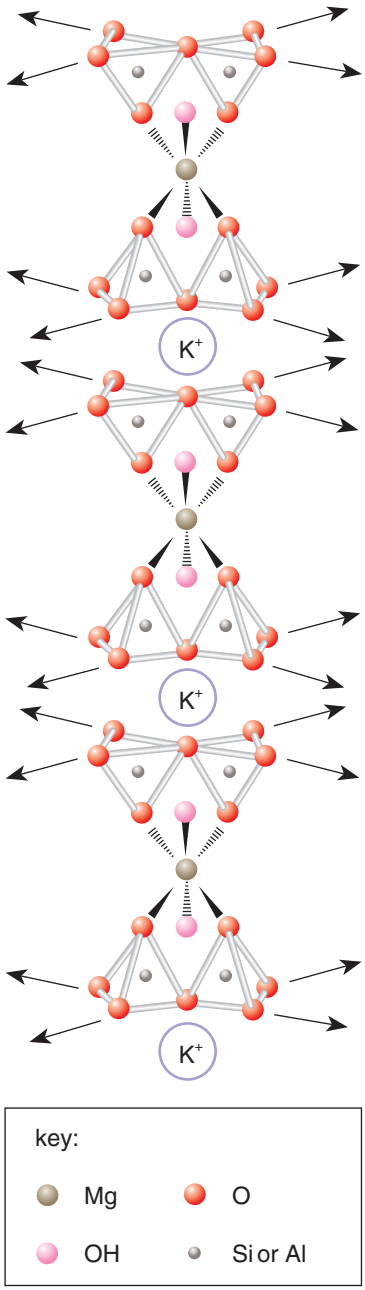


Fig. 49.27 The structure of mica; potassium ions are located between the layers

49.10 The aluminosilicates

Very often aluminium atoms can replace some of the silicon atoms at the centre of the tetrahedra in the silicates to form **aluminosilicates**. When an aluminium atom replaces a silicon atom, it contributes only three electrons to the bonding framework in place of the four electrons of a silicon atom. The remaining electron is supplied by the ionization of a metallic atom such as sodium or potassium.

The structure of mica minerals

Mica is a group of minerals. *Phlogopite*, $KMg_3(AlSi_3O_{10})(OH)_2$, is an example. In this mineral, one of every four silicon atoms is replaced by an aluminium atom*.

The replacement of one of every four silicon atoms by an aluminium atom requires a potassium atom to supply the fourth electron needed for electrical neutrality.

The mineral has a layer structure similar to that of talc. Fig. 49.27 shows the structure of mica. The potassium ions are located between the layers.

The attraction between the layers is increased by the presence of the extra ionic charge. The layers slide past one another less readily and this accounts for the hardness of mica over talc. Like talc, crystals of mica cleave easily into sheets (Fig. 49.28).



Fig. 49.28 A sample of mica, showing cleavage into a thin, transparent sheet

aluminosilicate 硅鋁酸鹽　　phlogopite 金雲母

Extension

Feldspars

Pure quartz is a continuous three-dimensional network of silicon and oxygen atoms. In feldspars, one or two out of every four silicon atoms is(are) replaced by aluminium atom(s).

Orthoclase, $KAlSi_3O_8$*, is a potassium feldspar. The silicon atoms and aluminium atoms occupy the centres of interlinked tetrahedra of SiO_4^{4-} and AlO_4^{5-}. These tetrahedra connect at each corner to other tetrahedra forming a three-dimensional, negatively charged framework. The potassium cations sit within the voids in this structure. N7



Fig. 49.29 Orthoclase

We have discussed natural polymers. We will look at synthetic polymers in the following sections.

* When one of every four silicon atoms is replaced by an aluminium atom, an anionic unit of the formula $AlSi_3O_8^-$ is obtained.

N7 Categories of feldspars

Feldspars are categorized by the cations they contain. Orthoclase, $KAlSi_3O_8$, is a potassium feldspar, albite, $NaAlSi_3O_8$, a sodium feldspar, and anorthite, $CaAl_2Si_2O_8$, a calcium feldspar*.

Feldspars may also contain both sodium and calcium ions, or sodium and potassium ions.

*In this calcium feldspar, two of every four silicon atoms are replaced by aluminium atoms.

49.11 Thermoplastics and thermosetting plastics

Most modern plastics are totally synthetic polymers. We can divide plastics into two broad groups based on their response to heating.

- Some plastics soften on heating. We can shape them and the new shape is retained on cooling. Such processes may be repeated. They are **thermoplastics**.
- Some plastics are hardened by heating. Subsequent heating does not soften them. They are **thermosetting plastics***.

* Thermoplastics are like chocolates. When you warm them they get soft. When cool they set hard. Thermosetting plastics are like eggs. Once you have hard boiled an egg, it sets and cannot be turned back to a liquid on further heating.

orthoclase 正長石 thermoplastic 熱塑性塑膠 thermosetting plastic 熱固性塑膠

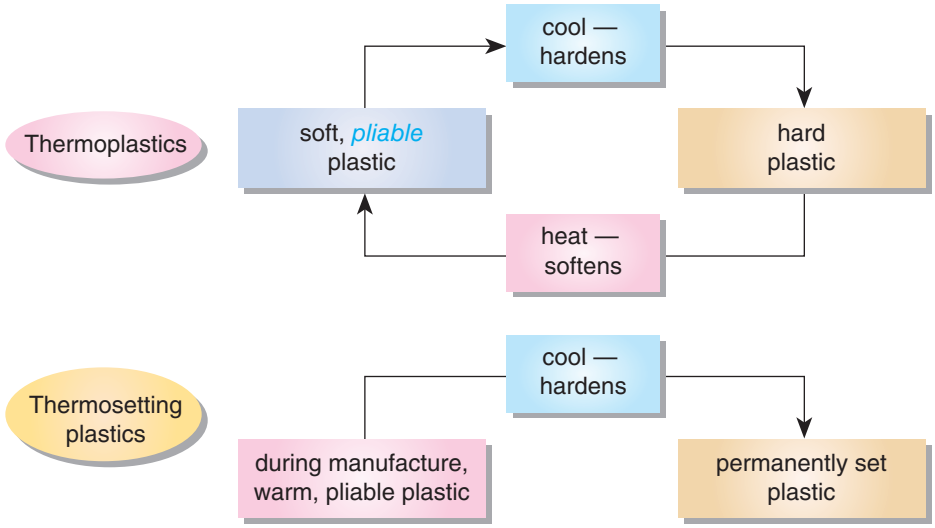


Fig. 49.30 The response of thermoplastics and thermosetting plastics to heating

* We have discussed these plastics in Topic 7 Fossil Fuels and Carbon Compounds.

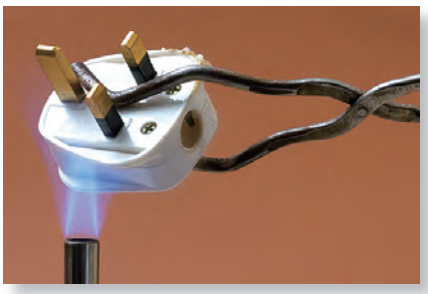


Fig. 49.31 Thermosetting plastics do not melt or soften on heating

ASL 2010 Paper 1 Q12
AL 2010 Paper 1 Q10

CE 2008 Paper 2 Q25

Thermoplastics can be melted or softened by heat at relatively low temperatures. They are flammable, flexible, strong and do not conduct electricity. They are also unreactive and inert to most substances. Polythene, polypropene, polyvinyl chloride, polystyrene and Perspex* are thermoplastics.

CE 2008 Paper 2 Q25

Thermosetting plastics do not melt or soften on heating. They are non-flammable, hard and rigid. They are also unreactive and inert to most substances, resistant to *creep* and deformation under a load and do not conduct electricity.

Fig. 49.32 shows some objects made of thermoplastics and thermosetting plastics.



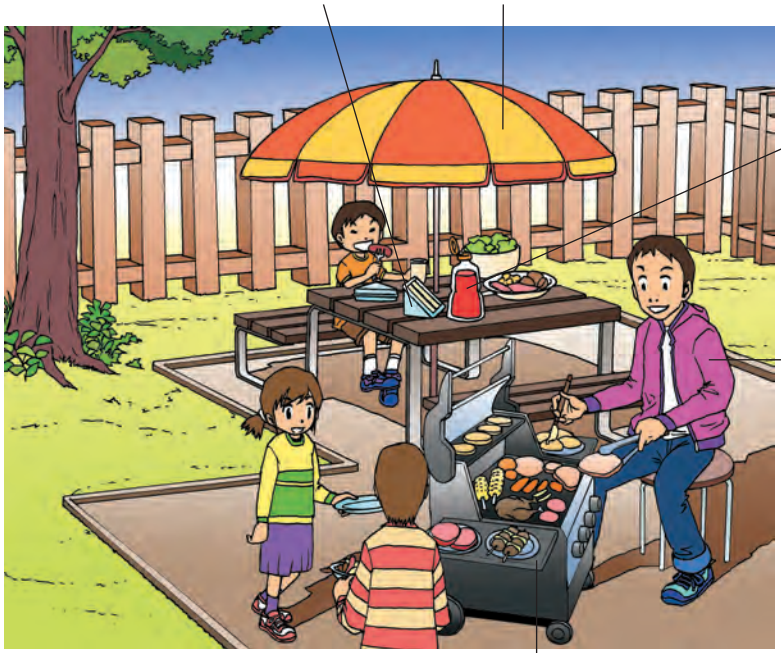
Fig. 49.32 Objects made of thermoplastics and thermosetting plastics

pliable 可屈折的　　creep 蠕變

Worksheet
A worksheet on this activity is available for download from *Jing Kung Chemistry Website*.

Discussion

Look at the garden shown below.



For each labelled item, suggest whether it is made of a thermoplastic or a thermosetting plastic.

49.12 Addition polymerization

ASL 2010 Paper 1 Q12
AL 2010 Paper 1 Q10

Polythene, polypropene, polyvinyl chloride, polystyrene and Perspex are addition polymers. They are formed by **addition polymerization**.

> Addition polymerization is a reaction in which monomer molecules join together repeatedly to form polymer molecules. No atoms are lost from the monomer molecules during the reaction.

In addition polymers, the monomer molecules usually contain C=C double bonds.

addition polymerization 加成聚合作用

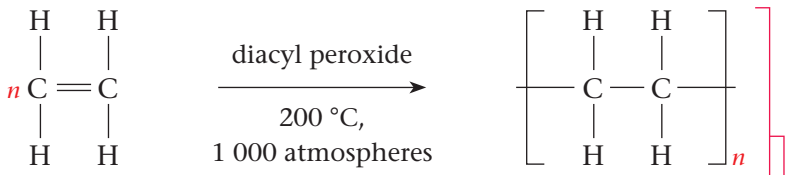
49.13 Production of polythene

Polythene is a thermoplastic. There are two main types of polythene: low density polythene (LDPE) and high density polythene (HDPE). The difference is due to the differences in production conditions.

Low density polythene

ASL 2012 Paper 2 Q7(a)
AL2012 Paper 2 Q11(a)

The addition polymerization of low density polythene is carried out at 200 °C and 1 000 atmospheres, in the presence of a small amount of an organic peroxide (called a diacyl peroxide).



ethene

low density polythene

CE 2006 Paper 1 Q11(b)(i)

Addition polymerization of ethene to produce low density polythene

Ethene is polymerized by free radical addition polymerization into low density polythene. The stages to focus on are

ASL 2002 Q12(b)(I)(ii)(2) ASL 2010 Paper 1 Q12 AL 2010 Paper 1 Q10

- initiation;
- propagation;
- termination.

AL 2001 Paper 2 Q5(c) AL 2003 Paper 2 Q5(b)(ii) AL 2006 Paper 2 Q7(b)(i)

Initiation

A diacyl peroxide molecule (RCOO–OOCR) undergoes a *homolytic fission* to give free radicals, which in turn initiate the chain reaction.

ASL 2002 Q12(b)(I)(ii)(3)



The radical (R•) produced reacts with an ethene molecule to form a new free radical.



homolytic fission 均裂作用

Reference Website
Refer to the following website for animations on free radical addition polymerization:
http://pslc.ws/macrog/radical.htm
(accessed May 2015)

Propagation

The resulting radical is electron-deficient and is very reactive. It can attack another ethene molecule to give a radical with a longer carbon chain.

$$RCH_2CH_2\bullet + CH_2CH_2 \longrightarrow RCH_2CH_2CH_2CH_2\bullet$$

By repeating the step, the carbon chain of the radical grows in length.

$$R(CH_2CH_2)_nCH_2CH_2\bullet + CH_2CH_2 \longrightarrow R(CH_2CH_2)_{n+1}CH_2CH_2\bullet$$

* During disproportionation, a free radical strips a hydrogen atom from another free radical so that one chain stabilizes with the hydrogen, and the other with a double bond.

Termination

In the termination step the radicals react to give a stable molecule, and the reaction stops.

$$2R(CH_2CH_2)_nCH_2CH_2\bullet \xrightarrow{\text{combination}} [R(CH_2CH_2)_nCH_2CH_2]_2$$

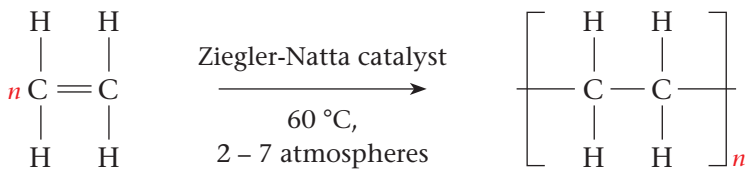
$$2R(CH_2CH_2)_nCH_2CH_2\bullet \xrightarrow{\text{disproportionation*}} R(CH_2CH_2)_nCH{=}CH_2 + R(CH_2CH_2)_nCH_2CH_3$$

Chain termination is a random process. Hence polythene will be made up of chains of different lengths.

High density polythene

* We have discussed Ziegler-Natta catalyst in Topic 12 Patterns in the Chemical World.

High density polythene is produced in a different way using catalysts called Ziegler-Natta catalysts*. Ziegler-Natta catalysts are mixtures of titanium compounds such as titanium(III) chloride ($TiCl_3$) or titanium(IV) chloride ($TiCl_4$), and compounds of aluminium like triethylaluminium ($(C_2H_5)_3Al$).



ethene — high density polythene

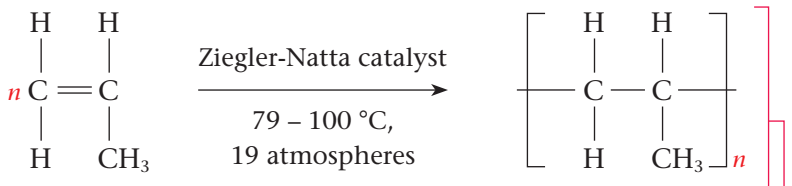
Extension

49.14 Production of polypropene, polyvinyl chloride and polystyrene

Besides polythene, we have discussed other addition polymers in Topic 7 Fossil Fuels and Carbon Compounds. We are going to look at the conditions needed for their production.

Polypropene

Polypropene is a thermoplastic. With the use of Ziegler-Natta catalyst (a mixture of titanium(III) chloride, magnesium chloride and ethyl benzoate), propene can be polymerized to polypropene.

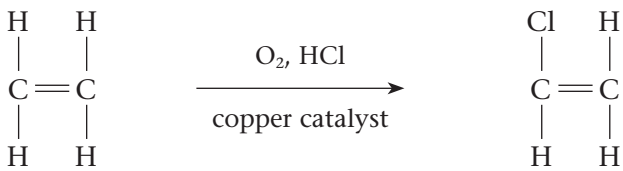


propene polypropene

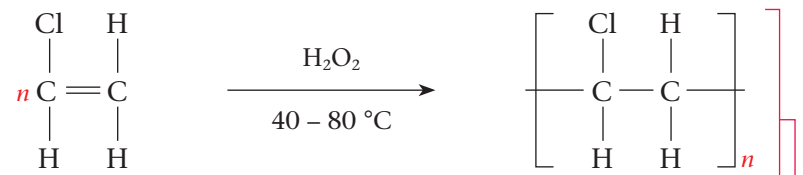
ASL 1999 Q7(I)(a)

Polyvinyl chloride

Polyvinyl chloride is also a thermoplastic. Vinyl chloride is the common name of chloroethene. It can be obtained by reacting ethene with oxygen and hydrogen chloride over a copper catalyst.



Vinyl chloride is polymerized by a free radical **mechanism** into polyvinyl chloride at a temperature of 40 – 80 °C in the presence of a peroxide initiator (e.g. hydrogen peroxide).



vinyl chloride polyvinyl chloride

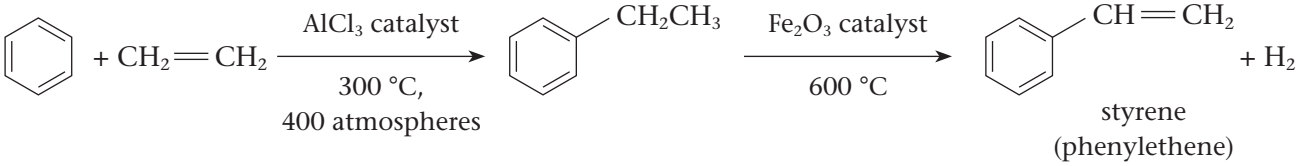
AL 1999 Paper 2 Q6(b)(i)(I)

mechanism 機理

Extension

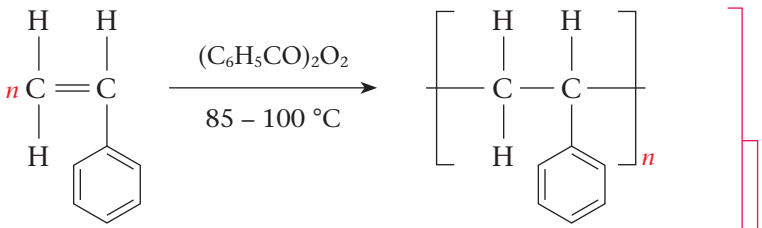
Polystyrene

Polystyrene is also a thermoplastic. Styrene is the common name of phenylethene. It is made from the reaction of benzene with ethene, followed by dehydrogenation.



styrene
(phenylethene)

The styrene produced is polymerized by a free radical mechanism into polystyrene at a temperature of 85 – 100 °C with dibenzoyl peroxide as the initiator.



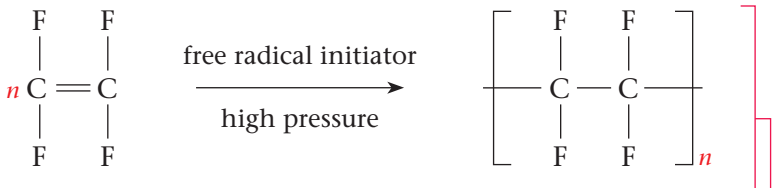
styrene polystyrene

ASL 2004 Q12(b)(II)(ii)(I)

In the following sections, we are going to look at some more addition polymers: **polytetrafluoroethene** (PTFE), **polymethyl methacrylate** (PMMA) and **poly(methyl cyanoacrylate)** (Superglue). They are also thermoplastics.

49.15 Polytetrafluoroethene (PTFE)

Tetrafluoroethene undergoes addition polymerization to give polytetrafluoroethene, a thermoplastic. Tetrafluoroethene is polymerized by a free radical mechanism under high pressure and in the presence of an initiator.



tetrafluoroethene polytetrafluoroethene

CE 2000 Paper 2 Q36
CE 2007 Paper 1 Q8(a)(ii)
AL 2004 Paper 1 Q4(a)(i)

polytetrafluoroethene 聚四氟乙烯 polymethyl methacrylate 聚甲基丙烯酸甲酯
poly(methyl cyanoacrylate) 聚氰基丙烯酸甲酯

Polytetrafluoroethene (PTFE) is often known to the public by the brand name *Teflon*. It contains very strong C–F bonds. These bonds make the polymer very stable and unreactive. PTFE is very resistant to attack by chemicals and is an excellent electrical insulator. It can be used up to 260 °C. N8

AL 2004 Paper 1 Q4(b)

The molecules of PTFE are surrounded by fluorine atoms. As fluorine is the most electronegative element*, an evenly distributed layer of partial negative charge is developed on the surface of a PTFE molecule (Fig. 49.33). This layer of fluorine atoms almost repels all other materials, thus preventing them from attaching to PTFE. This explains why PTFE has a slippery surface. It is used to coat frying pans to give them a non-sticking property.

* We have discussed electronegativity in Topic 6 Microscopic World II.

CE 2007 Paper 1 Q8(a)(i)

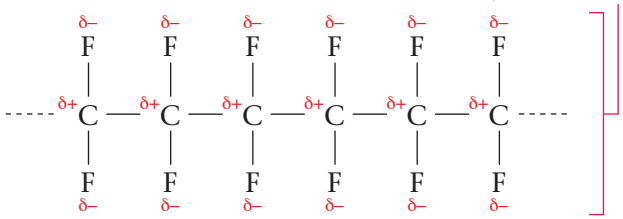


Fig. 49.33 An evenly distributed layer of partial negative charge on the surface of a PTFE molecule



Fig. 49.34 A non-stick frying pan

N8 Telflon — found by mistake

Roy Plunkett, an industrial chemist working for Du Pont, was cleaning out a cylinder of tetrafluoroethene gas when, to his surprise, he found that some of the gas had turned into a white slippery solid.

Further investigations into the curious substance found that it was resistant to heat and chemically unreactive. He could not get other substances to stick on it, as it had low surface friction. He realized that this substance was a polymer. Du Pont later named it Teflon.

In the 1950s, scientists discovered a method for bonding Teflon to aluminium. This increased sales of Teflon significantly and gave birth to the non-stick frying pan!

Other uses of polytetrafluoroethene

The roof of the Millennium Dome in England is made of glass fibre coated with PTFE, which is long lasting, easy to clean and non-flammable (Fig. 49.35).



Fig. 49.35 The Millennium Dome in Greenwich, England

Other uses of PTFE include cable insulation, bearings, gears, hoses and tubings.

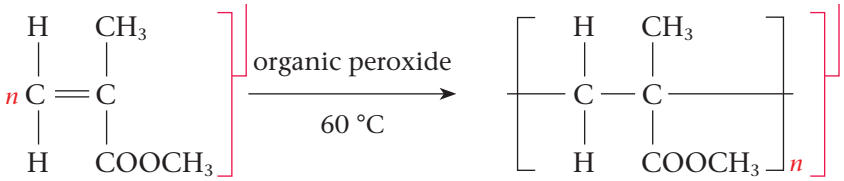
Teflon 特氟綸

49.16 Polymethyl methacrylate (PMMA)

Polymethyl methacrylate (PMMA) is produced by free radical addition polymerization of **methyl methacrylate** in the presence of an organic peroxide at about 60 °C.

AL 2005 Paper 2 Q7(c)(i)

CE 2000 Paper 2 Q36
ASL 2009 Paper 1 Q8(c)



methyl methacrylate
or
methyl 2-methylpropenoate

polymethyl methacrylate
or
poly(methyl 2-methylpropenoate)

Polymethyl methacrylate (PMMA) is often known by one of its trade names, for example in sheet form as Perspex and in powder form as Diakon. It is also commonly called acrylic glass or simply acrylic.

Polymethyl methacrylate (PMMA) is a thermoplastic. It is hard, strong, rigid and highly transparent. It has an excellent resistance to outdoor weathering. It is also highly scratch resistant.

Uses of polymethyl methacrylate

Some uses of PMMA are listed below.

- Cast sheet finds a major application in baths and other sanitary wares. It is also used as a lightweight replacement for glass.
- PMMA is used to make lenses of car lights.
- PMMA has a good degree of compatibility with human tissue. Special grades are used in applications such as false teeth and eyes and as a major component of bone cements.
- It is also used in adhesives, surface coatings and in paints*.

* Acrylic paint essentially consists of PMMA suspended in water.

methyl methacrylate 甲基丙烯酸甲酯



Fig. 49.36 Perspex products used in the laboratory



Fig. 49.37 Lenses of car lights are usually made of PMMA



Fig. 49.38 Acrylic painting

In the following activity, you are going to prepare Perspex.

Video
Preparing Perspex

Activity 49.2

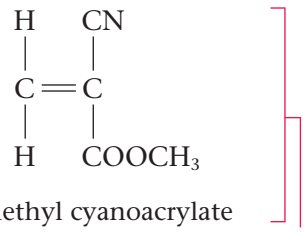
Preparing Perspex.

49.17 Cyanoacrylate (Superglue)

Methyl cyanoacrylate is the major ingredient of Superglue.



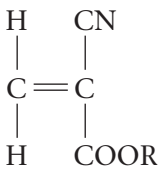
Fig. 49.39 Superglue



methyl cyanoacrylate

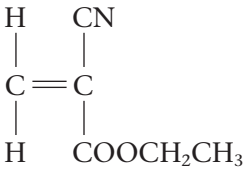
DSE 2013 Paper 2 Q2(b)(ii)(1) AL 2003 Paper 2 Q5(b)(i)

Methyl cyanoacrylate is just one member of a larger family of cyanoacrylates. Cyanoacrylates have the following basic structure:



The –R group is a variable which slightly changes the characteristic of the molecule.

Another member of the family, ethyl cyanoacrylate, is also commonly used in adhesive formulations.



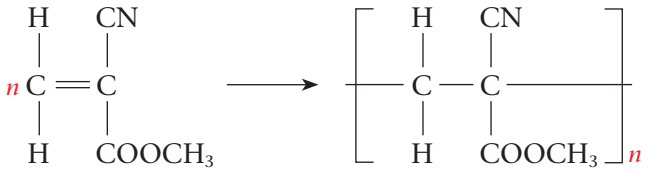
ethyl cyanoacrylate

CE 2008 Paper 1 Q8(a)

Superglue works via an addition polymerization. A trace of water* or hydroxide ions causes an extremely rapid polymerization of the cyanoacrylate monomer, forming a rigid plastic bonded to the surfaces on which it polymerized. This reaction happens within seconds. N9

* Even traces of water on our skin can cause the polymerization.

N9 In addition to putting back the cap for the Superglue that remains after use, keeping the Superglue in an air-tight container or in a dry place can help extend the lifetime of the Superglue.



methyl cyanoacrylate — poly(methyl cyanoacrylate)

CE 2008 Paper 1 Q8(b)

Because Superglue is so strong and bonds so fast, some people may accidentally glue two fingers together. We can dissolve the glue with propanone. There are also commercial removers available from glue manufacturers. N10

DSE 2013 Paper 2 Q2(b)(ii)(2)

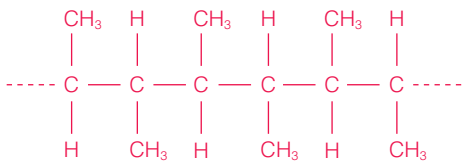
N10 Both propanone and poly(methyl cyanoacrylate) are polar organic substances. The same type of intermolecular forces exists in the two compounds. Thus, propanone can dissolve poly(methyl cyanoacrylate).

Medical grade superglues can be used to close wounds in place of stitches. N11

N11 Refer to page T22.

Checkpoint

1 Consider the addition polymerization of $CH_3CH{=}CHCH_3$. Draw the structure of the resulting polymer.



2 PMMA is often used in place of glass.

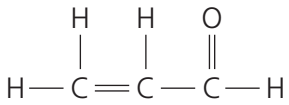
a) State TWO advantages of using PMMA instead glass.

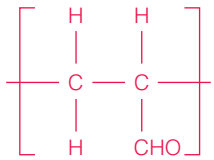
Lighter

Not easily broken

b) State TWO disadvantages of using PMMA instead glass.
More easily scratched
Does not filter ultraviolet light

3 The structure of acrolein is shown below:





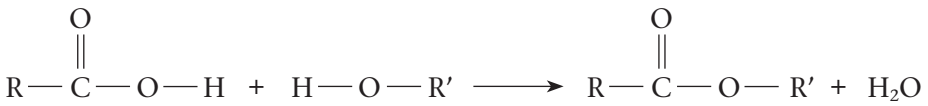
Acrolein undergoes addition polymerization readily. Draw the repeating unit of the polymer formed.

49.18 Condensation reaction and condensation polymerization

* We have discussed condensation reaction in Topic 8 Chemistry of Carbon Compounds.

A chemical reaction in which two or more molecules react together to form a larger molecule with the elimination of a small molecule is called a **condensation reaction***.

The reaction of a carboxylic acid and an alcohol to give an ester is a condensation reaction. Water is eliminated in the process.



carboxylic acid + alcohol → ester + water

The product molecules cannot react with other reactant molecules because they have no reactive functional groups left to react further. Hence no polymer forms. A special type of condensation reaction is **condensation polymerization**, in which a polymer forms.

condensation reaction 縮合反應　　condensation polymerization 縮合聚合作用

Condensation polymerization

In condensation polymerization, monomer molecules join together repeatedly as they do in addition polymerization. However, small molecules (usually H_2O, HCl or NH_3) are also formed.

ASL 2010 Paper 1 Q12
AL 2010 Paper 1 Q10

> Condensation polymerization is a reaction in which monomer molecules join together repeatedly to form polymer molecules. Small molecules, such as H_2O, HCl or NH_3, are always formed during the reaction. N12

CE 2002 Paper 1 Q8(c)(i)

In condensation polymerization, each monomer molecule must have two functional groups (or reactive sites) so that it can add on at each end to another unit of the chain. Condensation polymerization usually involves a reaction between two different monomers*.

* Condensation polymerization can also occur where a monomer molecule contains two different functional groups.

Many common polymers are made by condensation polymerization. They are called **condensation polymers**. Here we are going to look at some condensation polymers: **polyesters**, **polyamides** and **urea-methanal**.



Fig. 49.40 A collection of polyester threads

N12 Condensation polymerization — a step reaction

Condensation polymerization, also called a step reaction, has the features of a small reaction. Here two monomer molecules react together to give a dimer. A dimer then reacts with another monomer molecule or a dimer to produce a trimer or a tetramer. Thus, a higher molecular mass compound is gradually built up. Monomer disappears much faster in condensation polymerization as dimers, trimers and tetramers are formed at the same time. The molecular mass increases in the course of the reaction and high molecular mass polymer is obtained towards the end of the reaction. Long reaction time are, therefore, necessary for high percentage conversions and high molecular masses.

The scheme below shows the kinds of reactions that would lead eventually to a polymer:

monomer + monomer ⟶ dimer
dimer + monomer ⟶ trimer
dimer + dimer ⟶ tetramer
trimer + monomer ⟶ tetramer
trimer + dimer ⟶ pentamer
50-mer + 37-mer ⟶ 87-mer
etc. ⟶ polymer

49.19 Poly(ethylene terephthalate) (PET)

Polyesters are condensation polymers which contain the **ester linkage** $-\overset{\overset{O}{\|}}{C}-O-$. They are formed between dicarboxylic acids and diols.

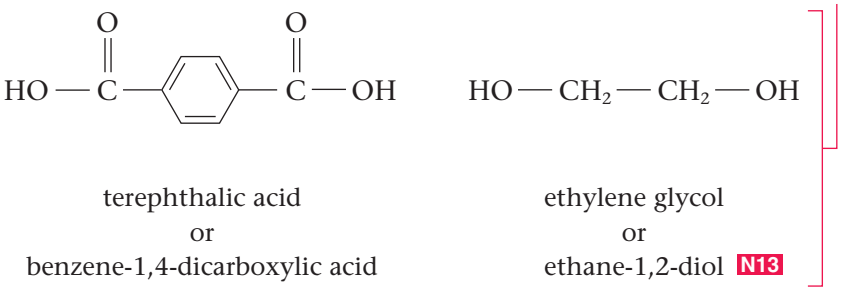
CE 2005 Paper 2 Q49
AL 1999 Paper 2 Q5(c)(ii)
AL 2002 Paper 2 Q7(c)(ii)

condensation polymers 縮合聚合物　polyester 聚酯　polyamide 聚酰胺
urea-methanal 脲甲醛　ester linkage 酯鍵合

* We have discussed PET in Topic 8 Chemistry of Carbon Compounds.

Poly(ethylene terephthalate) (PET)* is one of the most important polyesters. The names and structural formulae of the monomers for producing PET are shown below.

CE 2002 Paper 1 Q8(c)(ii) CE 2007 Paper 2 Q42 CE 2010 Paper 1 Q12(a)
ASL 2002 Q12(b)(I)(i) AL 1999 Paper 2 Q5(c)(ii)



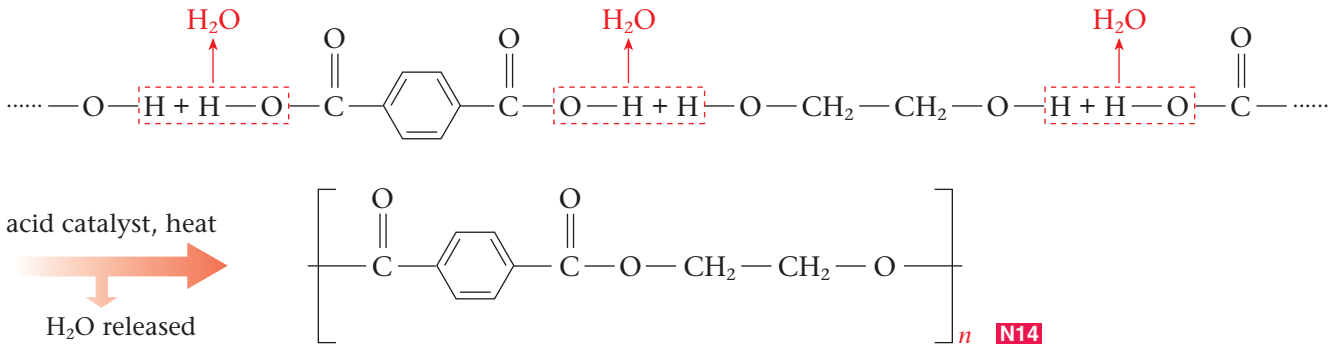
terephthalic acid
or
benzene-1,4-dicarboxylic acid

ethylene glycol
or
ethane-1,2-diol **N13**

N13 Examination questions often ask about structures of the monomers for producing PET.

Poly(ethylene terephthalate) can be obtained by a direct acid-catalyzed esterification of the two monomers. The acid and alcohol functional groups at both ends of the monomer molecules react to form ester linkages, releasing water molecules.

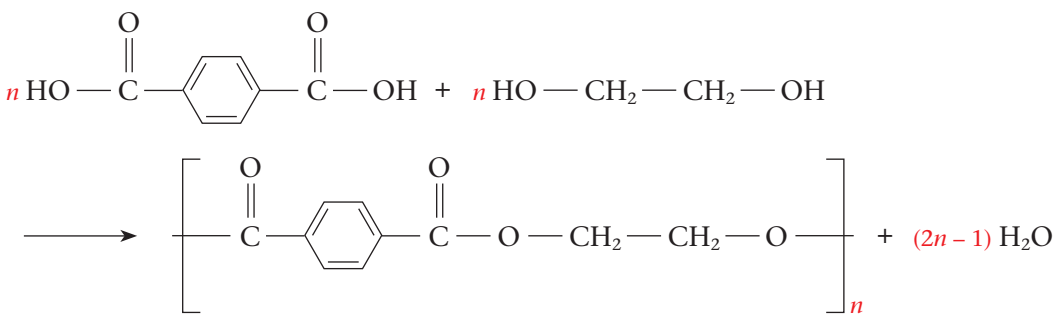
CE 2010 Paper 1 Q12(c)



N14

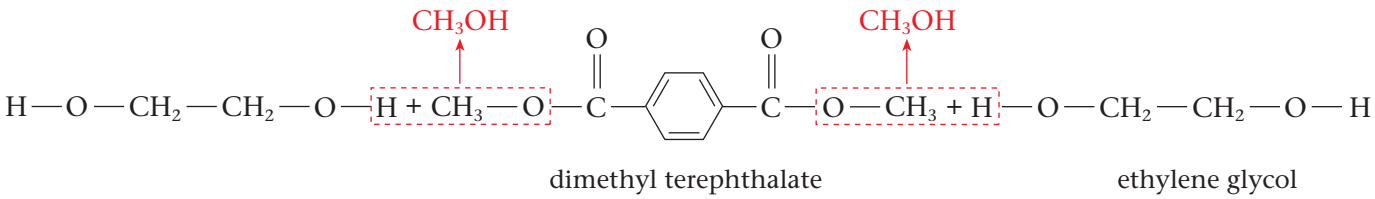
N14 The polymer chains of PET do NOT form hydrogen binds.

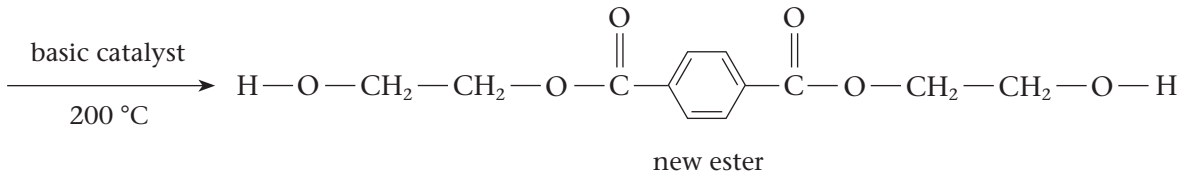
We can represent the condensation polymerization by the following equation:

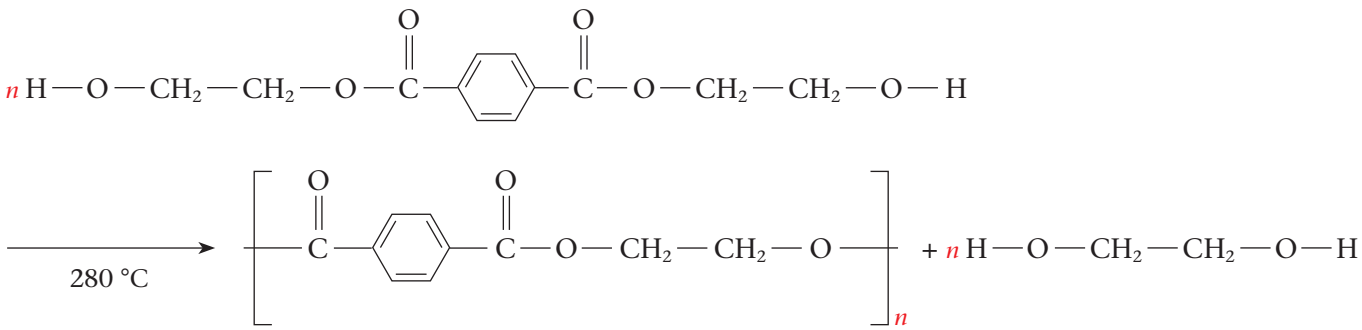


* poly(ethylene terephthalate) 聚對苯二甲酸乙二酯

One commercial synthesis utilizes **transesterifications** — reactions in which one ester is converted into another. In the first step, dimethyl terephthalate and excess ethylene glycol are heated to 200 °C in the presence of a basic catalyst. A new ester is formed. When this new ester is heated to 280 °C, ethylene glycol distills off and polymerization occurs.

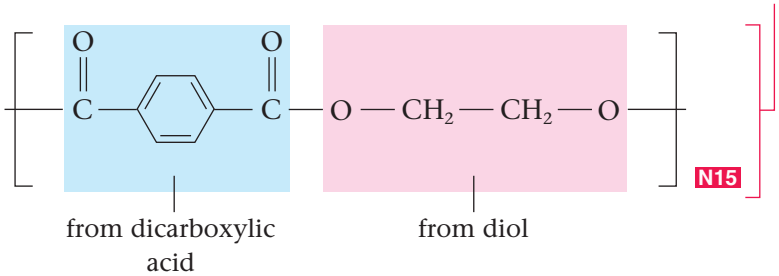






The repeating unit of poly(ethylene terephthalate) is

AL 2002 Paper 2 Q7(c)(i)



N15 The repeating unit of a polymer

Examination questions often give the repeating unit of a polymer and ask about information concerning the polymer. For example,

- whether it is an addition polymer or a condensation polymer;
- whether it is a thermoplastic or a thermosetting plastic;
- whether it is formed from one monomer or two different monomers;
- whether its polymer chains are cross-linked or not.

transesterification 酯基轉移作用

Uses of poly(ethylene terephthalate)

Terylene and Dacron for clothing

The fibres, *Terylene** and *Dacron*, are formed from the polyester by spinning. The fibres are strong and do not wrinkle easily. They are also resistant to stretching and shrinking, and quick drying. They are widely used to make clothes.

ASL 2010 Paper 1 Q12

* The name Terylene is derived from the names of the two monomers: terephthalic acid and ethylene glycol.



Fig. 49.41 Clothes made of Dacron



Fig. 49.42 Tapes and films made of PET

PET is also used for filling *anoraks* and *bedding duvets* to give good heat insulation. The polyester can also be made into films which are used in audio tapes and X-ray films.

For packaging

A newer use of PET is for packaging. The polyester has a great strength and a good chemical resistance. It also possesses good gas and moisture barrier properties. Hence PET is widely used for bottling carbonated drinks.

ASL 2010 Paper 1 Q12



Fig. 49.43 PET bottles for carbonated drinks

Worksheet
A worksheet on this activity is available for download from *Jing Kung Chemistry Website*.

Discussion

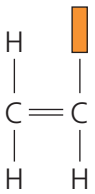
Compare the similarities and differences of addition polymerization and condensation polymerization. Present your results in table form.
Refer to page T23.

Terylene 特麗綾　Dacron 滌綸　anoraks 厚夾克　bedding duvets 羽絨被

Checkpoint

Refer to page T23.

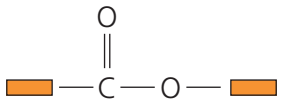
1 Study the structures of the following compounds:



I



II



III



IV



V

In the above structures, ▭ represents a saturated hydrocarbon chain containing 2 to 6 carbon atoms.

a) Which of the above compounds can be used to make an addition polymer? Draw the repeating unit of the polymer formed.

b) State a structural characteristic of monomers which can form addition polymers.

c) Which two compounds can be used to make a condensation polymer? Draw the repeating unit of the polymer formed.

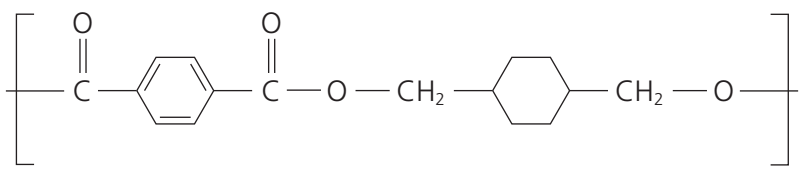
d) State a structural characteristic of monomers which can form condensation polymers.

2 The dimethyl ester of benzene-1,4-dicarboxylic acid also produces Terylene.

a) Draw the structural formula of this ester.

b) Predict the small molecule that is lost during the condensation polymerization with ethane-1,2-diol.

3 Kodel is another polyester that enjoys wide commercial use:



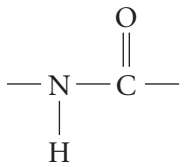
Kodel

a) Predict the structural formulae of the acid and alcohol that may be used to produce Kodel.

b) How can Kodel be degraded in the environment?

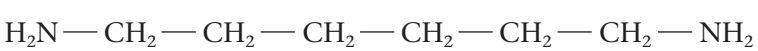
49.20 Nylons

* We have discussed nylons in Topic 8 Chemistry of Carbon Compounds.

Nylons* are well-known synthetic polymers. They are polyamides which contain the **amide linkage**  . They are made from diamines and dicarboxylic acids. N16

N16 Nylon is a thermoplastic because there is NO cross-link between its polymer chains.

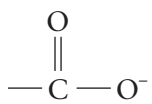
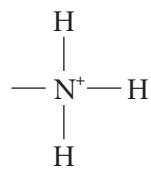
A common type of nylon is called nylon-6,6 because it is made from two monomers, each of which contains six carbon atoms in their molecules. The names and structural formulae of the monomers are shown below.



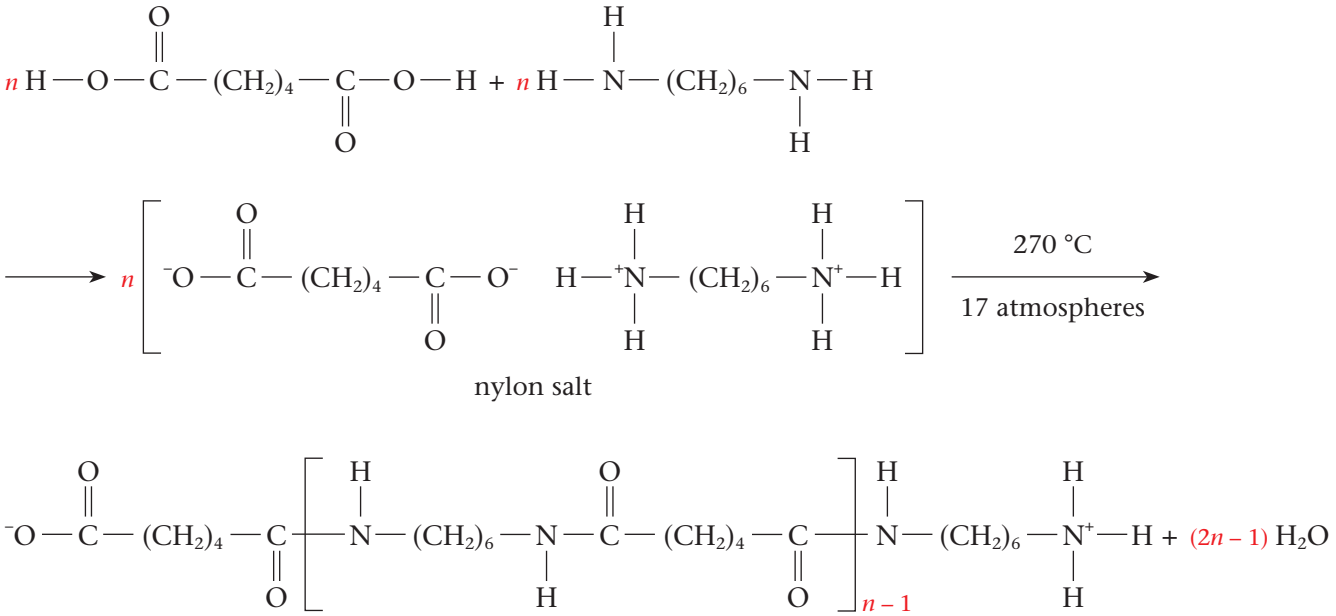
hexane-1,6-diamine



hexanedioic acid

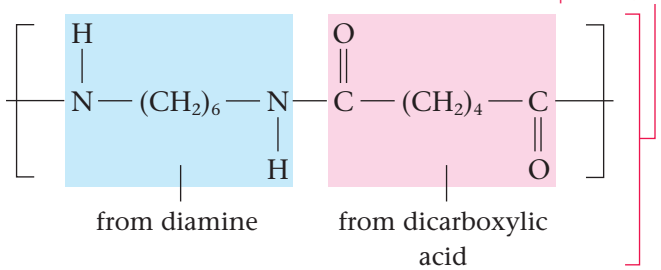
In the commercial process, these two monomers are allowed to react in equimolar proportions in order to produce a nylon salt. Then heating the nylon salt to 270 °C at a pressure of 17 atmospheres causes a polymerization to take place. Water molecules are lost as condensation reactions occurs between  and  groups of the salt to give the polyamide.

nylon 尼龍 amide linkage 酰胺鍵合



The repeating unit of nylon-6,6 can be written as

CE 2000 Paper 2 Q36
CE 2006 Paper 2 Q43



We describe different nylons by specifying the carbon atoms each monomer molecule contains (Table 49.2).

Table 49.2 Names of some common types of nylon

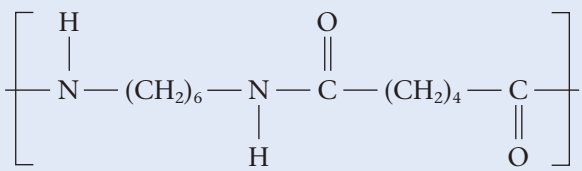
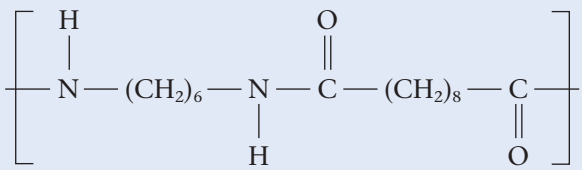
Name	Structural formulae of monomers		Repeating unit
	diamine	dicarboxylic acid	
Nylon-6,6	$H_2N-(CH_2)_6-NH_2$ (6 carbon atoms)	$HOOC-(CH_2)_4-COOH$ (6 carbon atoms)	
Nylon-6,10	$H_2N-(CH_2)_6-NH_2$ (6 carbon atoms)	$HOOC-(CH_2)_8-COOH$ (10 carbon atoms)	



Fig. 49.44 A nylon filament

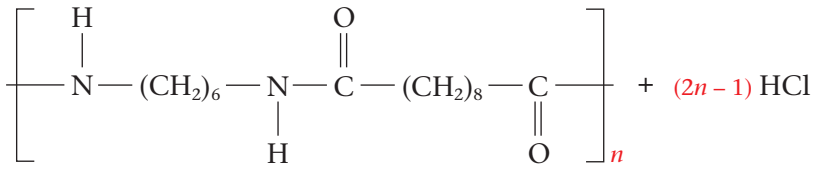
Video
Preparing nylon

Preparing nylon in the laboratory

The industrial preparation of nylon from a diamine and a dicarboxylic acid is quite slow. The reaction is faster when a chloride derivative of the acid is used.

$$n\,H_2N(CH_2)_6NH_2 + n\,ClCO(CH_2)_8COCl \longrightarrow$$

hexane-1,6-diamine decanedioyl dichloride



nylon-6,10

We can make nylon-6,10 in the laboratory by mixing aqueous solution of hexane-1,6-diamine in sodium carbonate solution with a solution of decanedioyl dichloride in cyclohexane (Fig. 49.45). Hydrogen chloride is eliminated during the process.

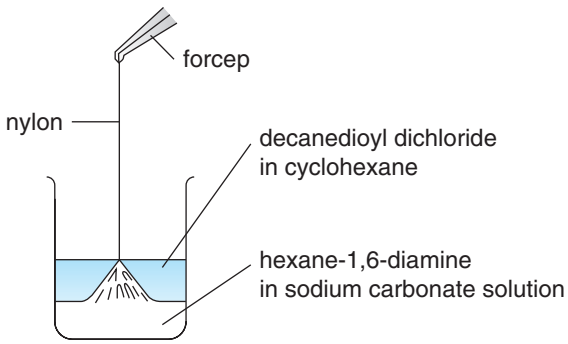


Fig. 49.45 Preparation of nylon in the laboratory



Fig. 49.46 Advertisement of nylon stocking when the polymer was first used

Uses of nylons

CE 2001 Paper 2 Q9

For clothing

One of the earliest uses of nylon was a substitute for silk in the manufacture of stockings.

Nylons have high melting points and a great strength. They are also *abrasion* resistant. They are extensively used as a textile fibre for clothes. Nylon fibres are also used in making ropes, fishing lines and nets.

abrasion 磨損

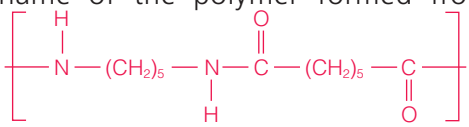
For machine parts

Nylons are used to replace steel parts in machines (e.g. gears and bearings). The usefulness of nylon arises from its excellent combination of strength, abrasion resistance and self-lubricating property, as well as its chemical unreactivity in many environments. They are also used in curtain rail fittings.

Checkpoint

1 Write the repeating unit and give the name of the polymer formed from the following monomers:

$H_2N(CH_2)_5NH_2$ and $HOOC(CH_2)_5COOH$



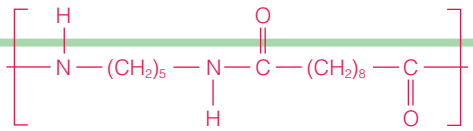
nylon-5,7

2 Nylon-5,10 can be formed by the reaction of a diamine with the chloride derivative of a dicarboxylic acid.

a) Write down the structures of the two monomers. $H_2N(CH_2)_5NH_2$ $ClCO(CH_2)_8COCl$

b) What small molecule is lost when the monomers react?
HCl

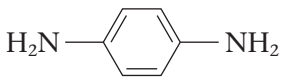
c) Draw the repeating unit of nylon-5,10.



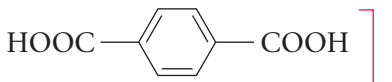
49.21 Kevlar

Kevlar is an aromatic polyamide. In the 1970s, one of the most significant achievements in the development of body armour was the invention of Du Pont's Kevlar — which is used as the padding in bullet-proof vests.

The names and structural formulae of the monomers of Kevlar are shown below:



1,4-diaminobenzene



benzene-1,4-dicarboxylic acid

DSE 2012 Paper 2 Q2(a)(i)(1)
ASL 2001 Q4(a)

Kevlar 凱庫勒

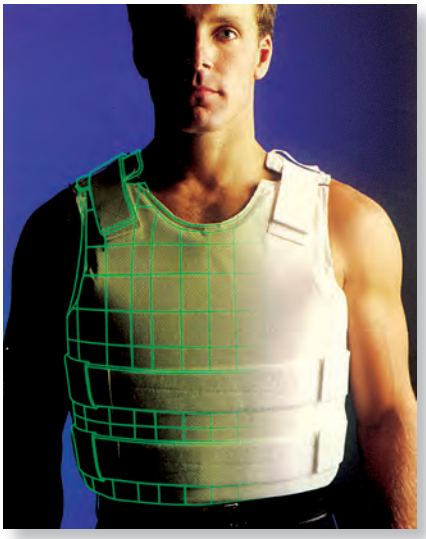
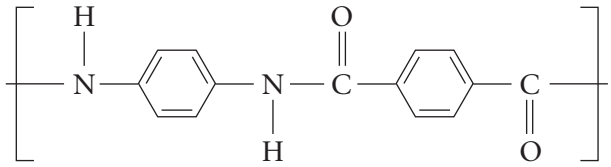


Fig. 49.47 A bullet-proof vest made from Kevlar

The repeating unit of Kevlar is



Benzene rings give the polymer chain more rigidity than an aliphatic chain.

Kevlar is a fibre which is fire-resistant, extremely strong* and flexible. It has a low density.

Uses of Kevlar

One of Kevlar's early uses was to replace steel cords in car tyres. Kevlar tyres are lighter and last longer than steel-reinforced tyres.

Kevlar ropes have 20 times the strength of steel ropes of the same weight. They last longer too. A stiffer form of Kevlar is used in aircraft wings, where its strength combined with its low density is important.

ASL 2010 Paper 1 Q12

* We will discuss the relationship between the structure of Kevlar and its strength in Unit 50.



Fig. 49.48 The yacht sail is made of Kevlar



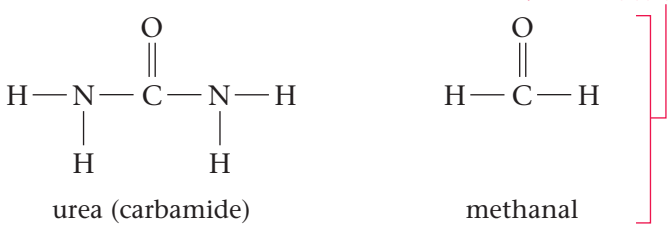
Fig. 49.49 A tennis racket with Kevlar frame and nylon strings

49.22 Urea-methanal

* We will compare the structures of thermoplastics and thermosetting plastics in Unit 50.

Urea-methanal is a thermosetting plastic* formed from the following two monomers:

CE 2007 Paper 2 Q42
AL 2006 Paper 2 Q6(c)(iv)



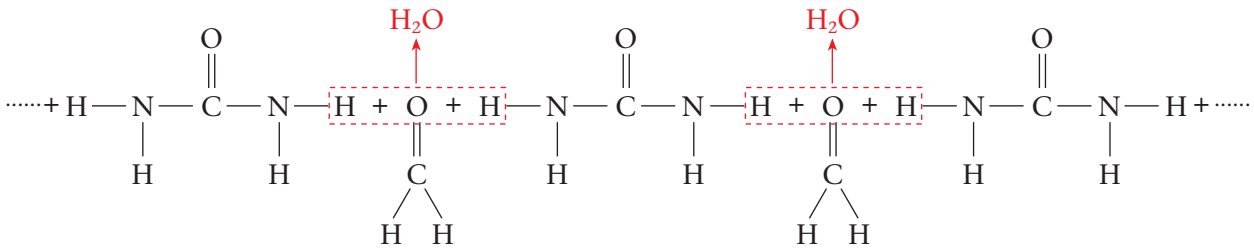
urea (carbamide) methanal

Heating urea and methanal in the presence of ammonia produces urea-methanal. There are two stages in this condensation polymerization:

- formation of long chain molecules;
- formation of **cross-links** between chains.

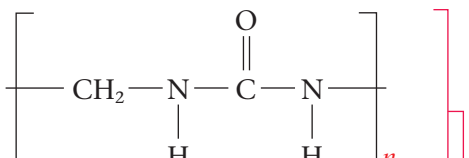
CE 2004 Paper 2 Q25

During the first stage, repeated condensation occurs with the elimination of water molecules, forming long chains.



urea methanal urea methanal urea

H_2O released — repeated condensation



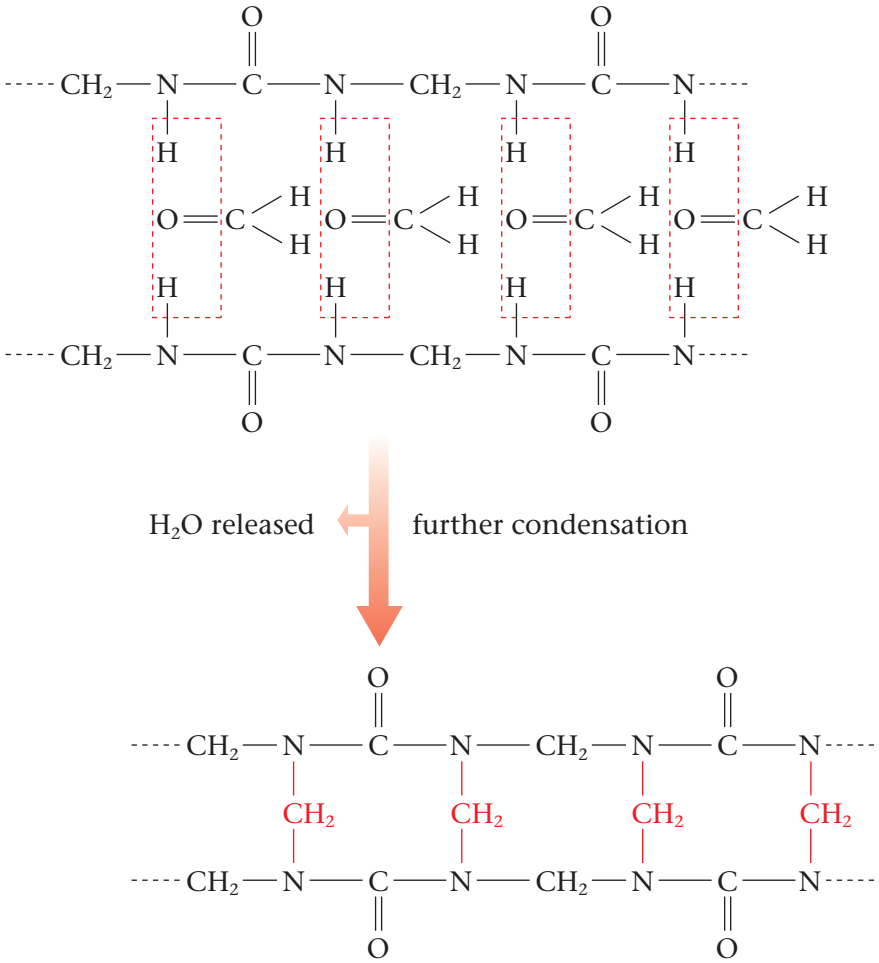
urea-methanal (without cross-links)

ASL 2012 Paper 1 Q5(a)
AL 2012 Paper 1 Q4(b)(i)

CE 2004 Paper 2 Q25

Upon further reaction, many cross-links (covalent bonds) form between the polymer chains. This results in a hard, rigid 3-dimensional network.

cross-link 交鍵



urea-methanal (with extensive cross-links)

Preparing urea-methanal in the laboratory

Follow the steps below when preparing urea-methanal in the laboratory:

1 Place 3 cm^3 of methanal solution in a test tube.

2 Add powdered urea to the solution in small amounts until some urea remain undissolved (Fig. 49.50a).

3 Pour the clear solution into a plastic vial (Fig. 49.50b).

4 Carefully add 1 drop of concentrated sulphuric acid to the solution (Fig. 49.50c).

5 Keep on stirring the mixture gently until a solid forms.

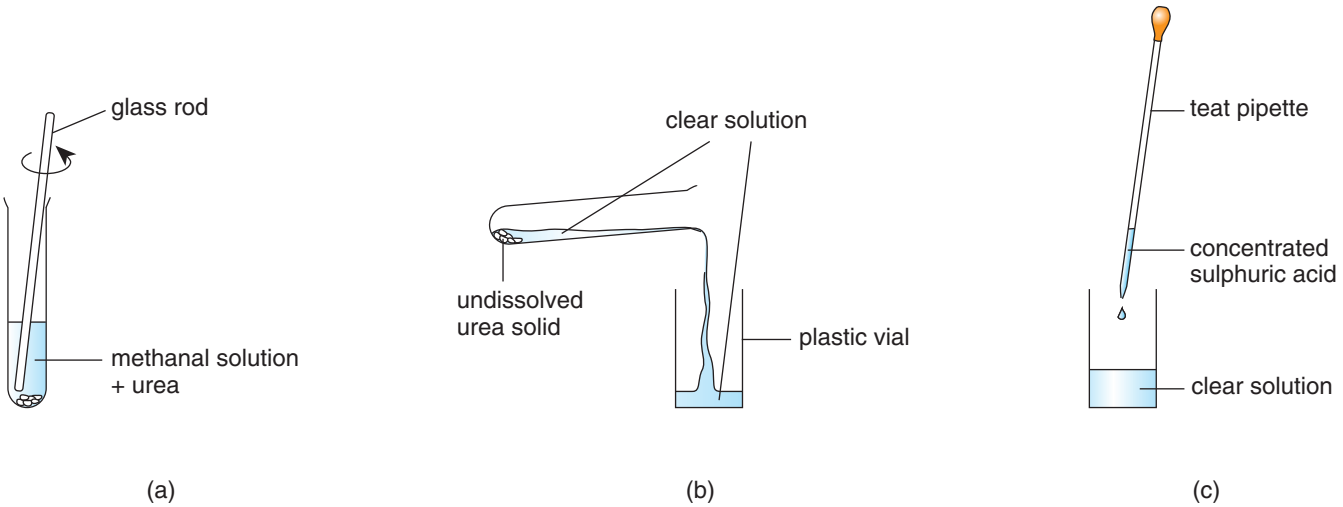


Fig. 49.50 Preparing urea-methanal in the laboratory

In the following activity, you are going to prepare two condensation polymers.

Video
Preparing urea-methanal

Activity 49.3

Preparing nylon and urea-methanal.

Uses of urea-methanal

ASL 2010 Paper 1 Q12

Urea-methanal is hard and strong. It is an excellent electrical insulator and also resistant to chemical attacks. Because of the cross-links, it cannot be softened or melted by heat and is insoluble in any solvent. Hence urea-methanal is mainly used in electrical fittings. It is also used in making handles of frying pans and control knobs. N17

N17 Urea-methanal wastes cannot be recycled by melting and remoulding.

DSE 2015 Paper 2 Q2(a)(ii)



Fig. 49.51 Items made of urea-methanal



Fig. 49.52 Control knobs made of urea-methanal

N18 Refer to page T24.

Table 49.3 summarizes the information about the plastics discussed above.

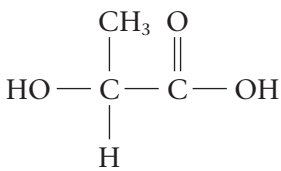
Table 49.3 Summary of information about plastics discussed

Plastic	Thermoplastic or thermosetting plastic	Addition polymer or condensation polymer N18
Polythene	thermoplastic	addition polymer — CE 1999 Paper 2 Q28
Polypropene	thermoplastic	addition polymer
Polyvinyl chloride	thermoplastic — DSE 2014 Paper 2 Q2(a)(iii)	addition polymer
Polystyrene	thermoplastic	addition polymer — CE 1999 Paper 2 Q28; DSE 2015 Paper 2 Q2(b)(ii)
Polytetrafluoroethene (PTFE)	CE 2008 Paper 2 Q41 — thermoplastic	addition polymer
Polymethyl methacrylate (PMMA)	thermoplastic	addition polymer
Poly(ethylene terephthalate) (PET)	CE 2005 Paper 2 Q49 — thermoplastic	CE 2008 Paper 2 Q41 — condensation polymer — CE 1999 Paper 2 Q28; CE 2006 Paper 2 Q43
Nylon-6,6	thermoplastic	condensation polymer
Kevlar	thermoplastic	CE 1999 Paper 2 Q28; CE 2008 Paper 2 Q41 — condensation polymer — CE 2008 Paper 2 Q41
Urea-methanal	DSE 2014 Paper 2 Q2(a)(iii) — thermosetting plastic	condensation polymer

49.23 Polylactide (PLA) — a polymeric biomaterial

DSE 2014 Paper 2 Q2(c)(iv)(1)

Polylactide* (PLA) is a polyester whose monomer is lactic acid.



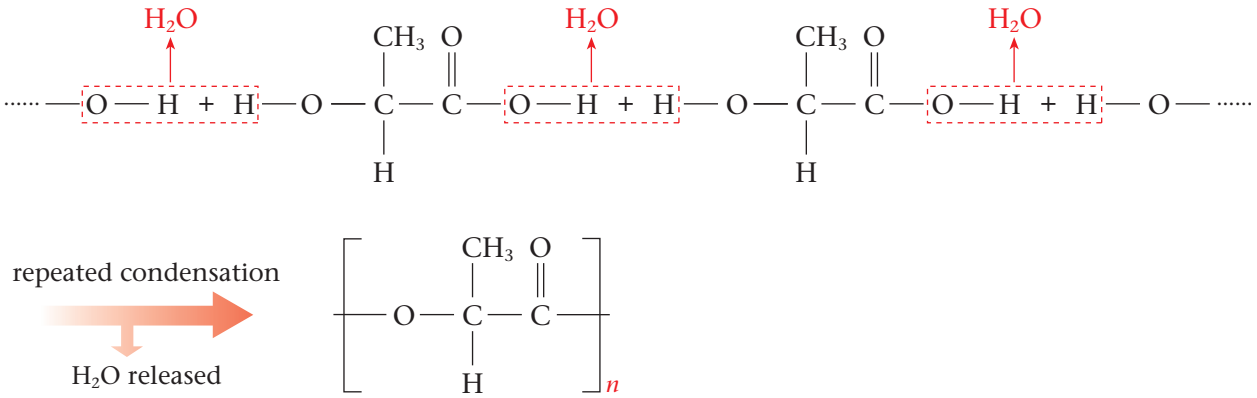
lactic acid (2-hydroxypropanoic acid)

* We will discuss polylactide further in Unit 50.

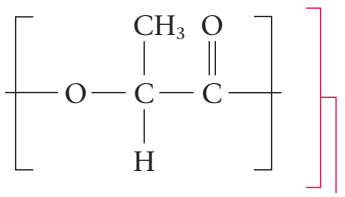
polylactide 聚乳酸

Lactic acid occurs naturally in animals and microorganisms. It is found in many natural foods and in fermented foods such as yogurt.

Polylactide is a polymeric biomaterial as it is polymerized from lactic acid, a natural biomaterial. Lactic acid undergoes condensation polymerization to form polylactide.

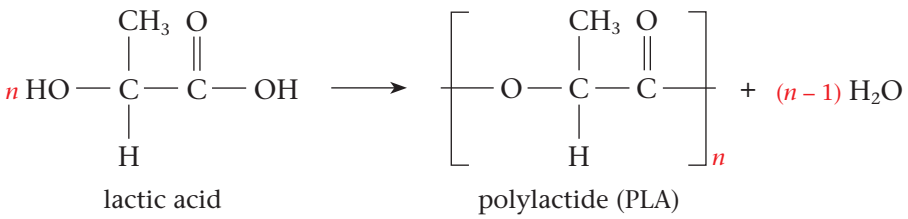


The repeating unit of polylactide is



CE 2006 Paper 1 Q11(c)(ii)

We can represent the condensation polymerization by the following equation:



Polylactide is a thermoplastic.

Worksheet Assessment Form

A worksheet and an assessment form on this activity are available for download from *Jing Kung Chemistry Website*.

Internet Search & Presentation

Polymeric materials used as adhesives and drug-carriers

Task

Your teacher will divide the class into groups. Each group should give a 4-minute presentation in one area of usage of polymeric materials:

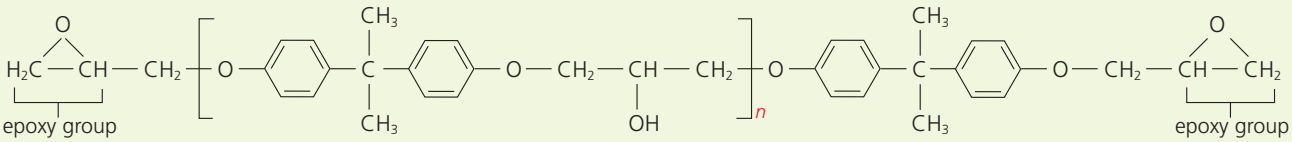
- as adhesives, or
- as drug-carriers.

Support your presentation with various aids.

Use of polymeric materials as adhesives

There are many advantages that polymeric adhesives can offer compared with the more traditional methods of joining such as bolting and welding. Epoxy adhesives represent the most common type of structural adhesive.

Epoxy adhesives usually have two components that are mixed just before use. The first part is a low molecular mass polymer with epoxy groups at each end. Its structure looks like this:



The second part is a diamine. When you mix the two components together, they react to give a cross-linked network.

Hints for the search Refer to page T27.

1 How do the two components of an epoxy adhesive react to give a cross-linked network?

2 How can the toughness of epoxy adhesives be increased?

Reference websites

1 You may start your search from the following websites:

- Epoxy resins — from Macrogalleria, a website on polymer
  http://pslc.ws/macrog/epoxy.htm

- Toughening epoxy adhesives to meet today's challenges
  http://uksaa-www.me.ic.ac.uk/MRS.v5.pdf.

2 Extend your search using search engines and appropriate keywords.

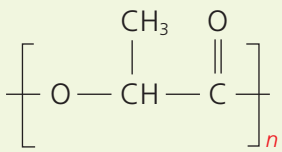
- http://www.zoo.com/Zoo-Site/
- http://www.google.com
- http://www.yahoo.com
- http://www.searchalta.com/

Use of polymeric materials as drug-carriers

Till early 1970, drugs were delivered to the human body exclusively via oral and *intravenous* means. In 1970, Yolles and coworkers reported the use of lactic acid based polymers for drug delivery. Since then a lot of effort has gone into developing new methods of drug delivery.

Controlled drug delivery occurs when a polymer is combined with a drug in such a way that the drug is released from the material in a pre-designed manner. One possible method is to plant a polymeric implant, which consists of the drug embedded in a polymer matrix, into the affected area of the body. The drug is then released directly into the affected site via diffusion or surface erosion.

The selection of an appropriate polymer matrix is essential for the development of a successful drug delivery system. Polylactide (PLA) is commonly used for this application. The drug is released at a rate determined by the rate of hydrolysis of the polyester.



PLA
polylactide (PLA)

Hints for the search Refer to page T27.

1 Suggest properties that an 'ideal' polymer for drug delivery should possess.

2 Discuss the use of PLA in controlled drug delivery.

Reference websites

1 You may start your search from the following websites.

- A brief review on polymeric drug delivery
  http://www.drugdel.com/polymer.htm

intravenous 靜脈內的

- An overview on drug delivery with emphasis on polymeric implants
  http://chaubal.tripod.com/review.html
- An article on the use of biodegradable polymers in drug delivery systems to provide pre-programmed release
  http://cdn.intechopen.com/pdfs-wm/46807.pdf

2 Extend your search using search engines and appropriate keywords.

- http://www.zoo.com/Zoo-Site/
- http://www.google.com
- http://www.yahoo.com
- http://www.searchalta.com/

Worksheet

Two related worksheets with answers are available for download from *Jing Kung Chemistry Website*. Worksheet ① consists of additional questions for promoting active reading. (Note: Questions in this worksheet also cover those in the textbook.) Worksheet ②, requiring students to indicate the relationship among various terms by drawing, may help the less able readers improve their reading skills.

STSE Connections

Gore-Tex

Breathable, yet waterproof. When it comes to fabric these two qualities would seem to be at odds with one another and indeed they were, until 1969. That was when the father-son team of Bill and Robert Gore developed the miraculous product known as **Gore-Tex**.



Fig. 49.53 A trademark for Gore-Tex

Structure of the Gore-Tex membrane

In 1969, the Gore team discovered that polytetrafluoroethene (PTFE) could be stretched to form a porous material which would allow water vapour but not water liquid to pass through the minute holes.

The Gore team made use of this property of PTFE to develop a material — Gore-Tex. Gore-Tex is a bi-component membrane. The main part is made from expanded polytetrafluoroethene (ePTFE). ePTFE is formed into a film containing billions of pores per cm^2. Each pore is 20 000 times smaller than a raindrop, but 700 times larger than a molecule of water. It is completely impervious to rain and snow, but allows perspiration vapour to pass through. This is then combined with an oil-hating polymer layer. This layer allows the water vapour through, but prevents the natural oil from the skin blocking the pores in the ePTFE, thus preserving its properties. The result is a totally waterproof, windproof and breathable membrane.

Gore-Tex 戈爾特斯

There are a number of different Gore-Tex constructions currently used within the industry. The classic 2-layer construction is made by sandwiching the Gore-Tex membrane between two layers of fabric.

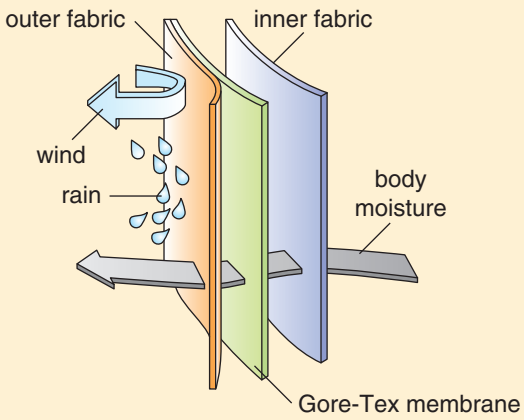


Fig. 49.54 How a Gore-Tex membrane works

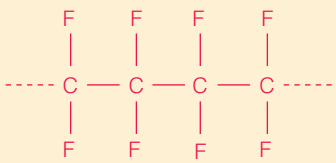


Fig. 49.55 Gore-Tex clothing is ideal for outdoor activities

The outer face of the Gore-Tex fabric is coated with a hydrophobic DWR (Durable Water Repellency) treatment. This encourages surface water to bead up and run off, improving the wet weather performance of the garment and promoting breathability by preventing the wetting-out of the outer face.

Questions

1 What is the structure of polytetrafluoroethene (PTFE)?



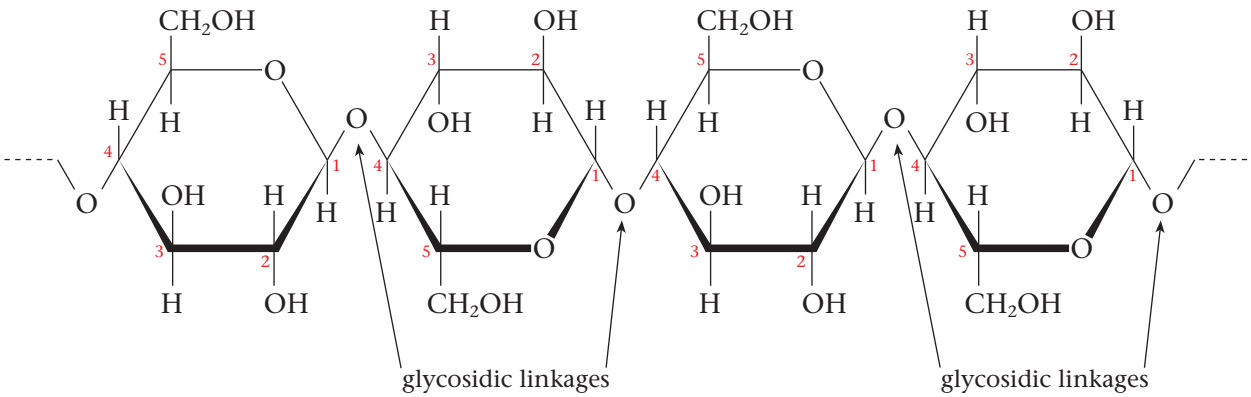
2 What are the structural features of Gore-Tex?
It is impervious to rain and snow, but allows perspiration vapour to pass through.

3 Gore-Tex is used in a wide variety of applications besides water-proof clothing. Search for other uses of Gore-Tex. GORE-TEX boots and shoes – Our foot perspires roughly a full cup of moisture a day during activity. GORE-TEX boots and shoes are both waterproof and breathable, keeping external moisture out while allowing moisture vapour from perspiration to escape.
GORE-TEX caps and hats – Cover the head so that moisture stays out and sweat is not locked in.

Summary

1 Materials can be divided into four classes: metals, polymers, ceramics and composites. Polymers may be natural or synthetic.

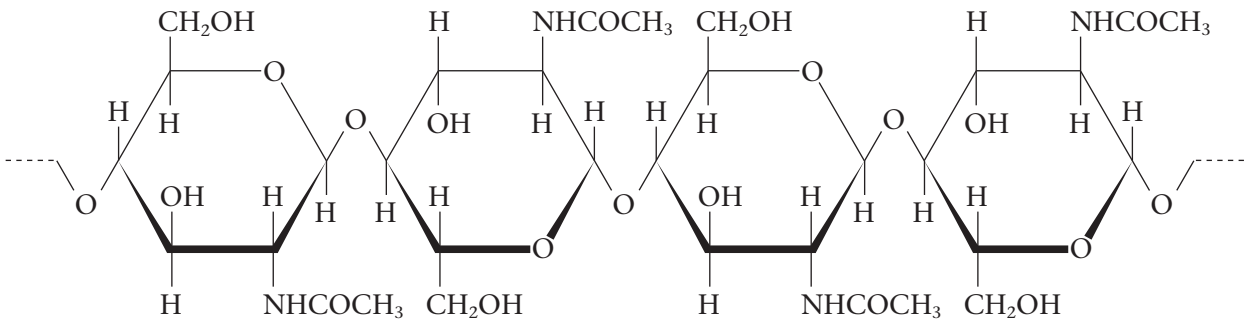
2 a) Cellulose is a natural polymer of β-glucose monomer.



b) The chains of cellulose have a linear structure. Hydrogen bonds exist between the chains. The alignment of the chains and the hydrogen bonds between them are responsible for the high strength of cellulose.

c) Cellulose is a biodegradable polymer.

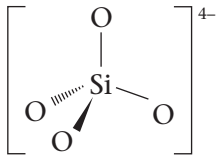
3 a) Chitin is the Earth's second most abundant carbohydrate polymer.



b) The chains of chitin are packed side by side and are held together by hydrogen bonds. Hence it is hard and insoluble in water.

Extension

4 a) All silicate minerals are built up from the basic unit of orthosilicate anion, SiO_4^{4-}.



b) The following table summarizes some important information about silicate minerals.

Nature of structure	Single chain	Double chain	Sheet	Network
Number of corners shared for each tetrahedron	2	2.5	3	4
Representation for the overall structure	$(SiO_3^{2-})_n$	$(Si_4O_{11}^{6-})_n$	$(Si_2O_5^{2-})_n$	$(SiO_2)_n$
Example	pyroxenes, $CaMg(Si_2O_6)$	amphibole, $Ca_2Mg_5(Si_4O_{11})_2(OH)_2$	talc, $Mg_3(Si_4O_{10})(OH)_2$	quartz, $(SiO_2)_n$

5 a) Aluminosilicates is a group of minerals in which the aluminium atoms replace some of the silicon atoms at the centre of the tetrahedron in the silicates.

b) Mica and feldspars are aluminosilicates.

6 Plastics can be classified into:

a) thermoplastics — can be softened on heating;

b) thermosetting plastics — do not soften on heating.

7

Polymerization

the process of repeatedly joining together many small molecules to form very large molecules

Addition polymerization

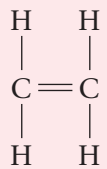
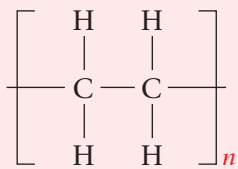
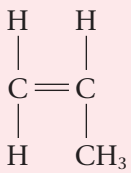
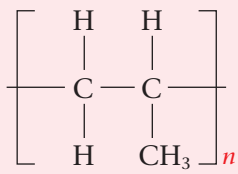
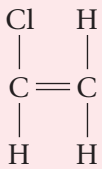
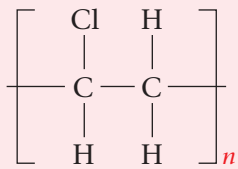
- a reaction in which monomer molecules join together repeatedly to form polymer molecules
- no atoms are lost from the monomer molecules during the reaction
- monomer molecules usually contain C=C double bonds

Condensation polymerization

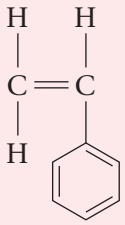
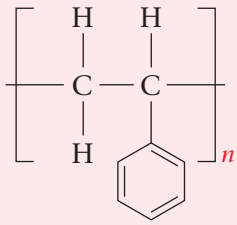
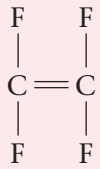
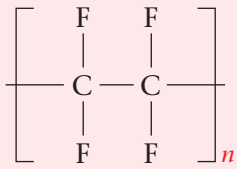
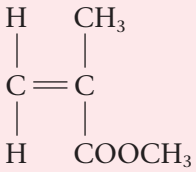
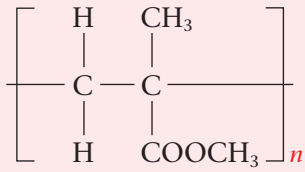
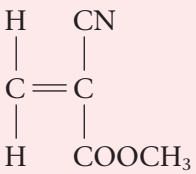
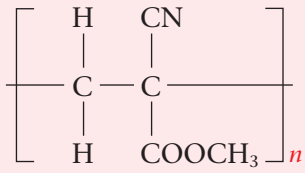
- a reaction in which monomer molecules join together repeatedly to form polymer molecules
- small molecules are formed during the reaction
- each monomer molecule must have two functional groups (or reactive sites)
- usually involves a reaction between two different monomers

Summary

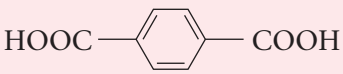
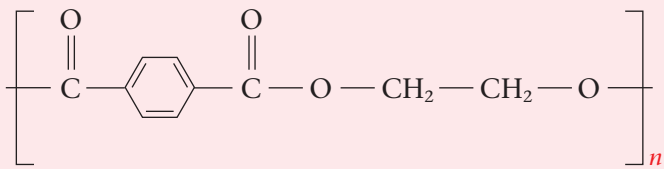
8 The table below gives the information about some addition polymers. All are thermoplastics.

Name	Monomer	Structure of polymer
Polythene		
Polypropene		
Polyvinyl chloride		

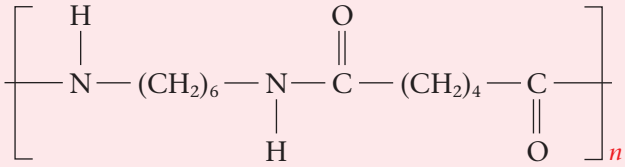
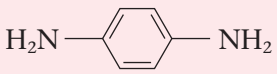
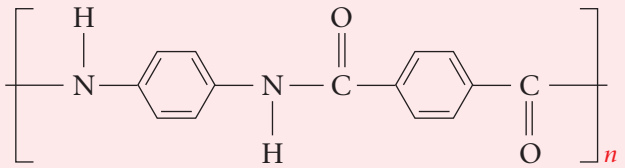
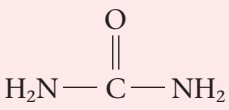
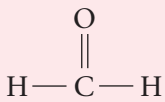
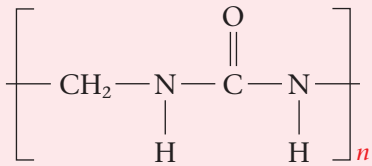
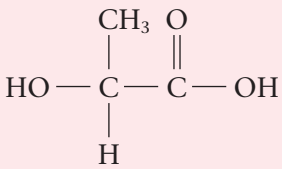
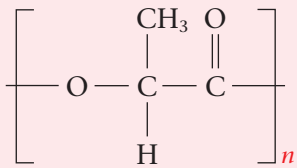
Continued on next page

Name	Monomer	Structure of polymer
Polystyrene		
Polytetrafluoroethene		
Polymethyl methacrylate (PMMA)		
Poly(methyl cyanoacrylate) (Superglue)		

9 The following table gives the information about some condensation polymers.

Name	Monomers	Structure of polymer
Poly(ethylene terephthalate) (a thermoplastic)	 terephthalic acid  ethylene glycol	

Continued on next page

Name	Monomers	Structure of polymer
Nylon-6,6 (a thermoplastic)	$H_2N-(CH_2)_6-NH_2$ hexane-1,6-diamine $HOOC-(CH_2)_4-COOH$ hexanedioic acid	
Kevlar (a thermoplastic)	 1,4-diaminobenzene  benzene-1,4-dicarboxylic acid	
Urea-methanal (a thermosetting plastic)	 urea  methanal	
Polylactide (PLA) (a polymeric biomaterial) (a thermoplastic)	 lactic acid	

Exercise

Note: The symbol 🌶 indicates the level of difficulty of a question.

Part I Knowledge and understanding

1 Construct a concept map using the following concept words. You may also add other concept words to your concept map.
Refer to page T30.

natural polymers, synthetic polymers, thermoplastics, thermosetting plastics, addition polymerization, condensation polymerization, cellulose, chitin, polythene, polypropene, polyvinyl chloride, polystyrene, polytetrafluoroethene, polymethyl methacrylate, poly(methyl cyanoacrylate), polylactide, poly(ethylene terephthalate), nylon, Kevlar, urea-methanal

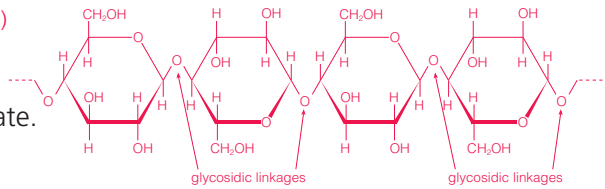
2 Cellulose is a natural polymer.

a) Name the monomer that polymerizes to form cellulose.
β-glucose

b) When two monomer units combine to form a dimer, name the linkage that joins these two units.
Glycosidic linkage

c) Draw a section of a cellulose chain.

c)



3 Chitin is the Earth's second most abundant carbohydrate.
Refer to page T31.

a) Suggest where chitin can be found in animals.

b) Draw the structure, showing two repeating units, of chitin.

4 Describe the differences between
Refer to page T31.

a) thermoplastics and thermosetting plastics;

b) addition polymerization and condensation polymerization.

5 Plastics can be classified in the following ways:

- depending on how they response to heating — thermoplastics and thermosetting plastics
- depending on the method of polymerization — addition polymers and condensation polymers

Classify the following plastics according to the above ways.

Plastic	Thermoplastic or thermosetting plastic	Addition polymer or condensation polymer
Polypropene (PP)	thermoplastic	addition polymer
Polyvinyl chloride (PVC)	thermoplastic	addition polymer
Polytetrafluoroethene (PTFE)	thermoplastic	addition polymer
Polymethyl methacrylate (PMMA)	thermoplastic	addition polymer
Poly(ethylene terephthalate) (PET)	thermoplastic	condensation polymer
Nylon-6,6	thermoplastic	condensation polymer
Kevlar	thermoplastic	condensation polymer
Urea-methanal	thermosetting plastic	condensation polymer

Exercise

6 Complete the following table.
Refer to page T31.

Polymer	Monomer(s)	Structure of polymer
Polytetrafluoroethene		
Polymethyl methacrylate		
Poly(methyl cyanoacrylate)		
Polylactide		
Poly(ethylene terephthalate)		
Nylon-6,6		
Kevlar		
Urea-methanal		

7 b) Polypropene can be disposed of by incineration.
It is not appropriate to use landfilling because polypropene is non-biodegradable.
Cellulose can be disposed of by landfilling.
Cellulose is biodegradable. It can be decomposed by microorganisms.

Part II Structured questions

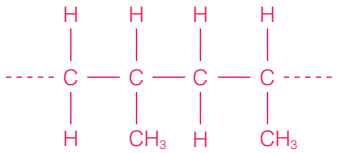
Answers for the questions of the public examinations in Hong Kong are not provided.

7 a) Draw the structure, showing two repeating units, of each of the following polymeric materials:

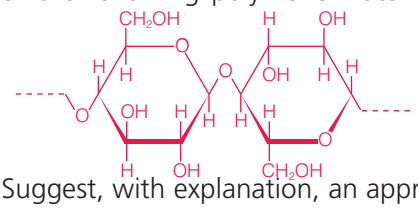
i) polypropene; and

ii) cellulose.

i)



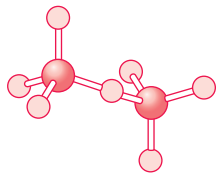
ii)



b) Large quantities of polypropene and cellulose are used daily. Suggest, with explanation, an appropriate waste disposal method for each material.

Extension

8 Draw the structure of the anionic part of the mineral akermanite, $Ca_2Mg(Si_2O_7)$.



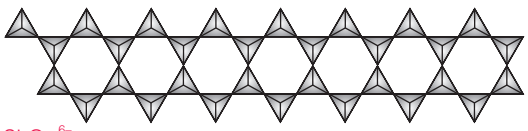
9 Determine the repeating unit of each of the following silicates.

a)



SiO_3^{2-}

b)



$Si_4O_{11}^{6-}$

key:

orthosilicate anion

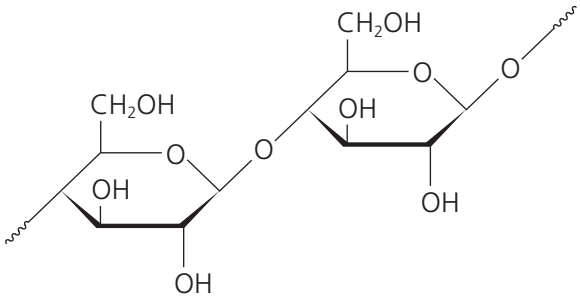
10 a) Explain why asbestos has a fibrous structure.

b) Explain why talc can be used as a major ingredient in baby powder.

10 a) Asbestos has a double chain structure. The electrostatic attraction between the cations and the negative silicate chains holds the structure together.
Crystal cleavage can occur along the chain direction because the non-directional ionic links are weaker than the strong Si–O covalent bonds. Hence the asbestos has a fibrous structure.

b) Talc has a layer structure. The layers are held together by weak van der Waals' forces. The layers can slide past one another easily.

11 A part of the structure of cellulose is shown below:



Cellulose is a condensation polymer of glucose.

a) What is the meaning of the term 'condensation polymer'?

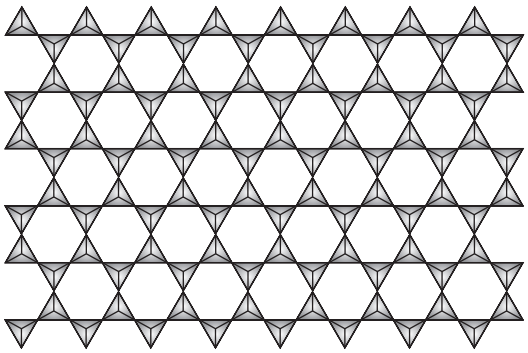
b) Draw the structure of a molecule of glucose.

(HKDSE 2013)

Exercise

Extension

12 The following diagram shows a portion of a silicate sheet.



key:
orthosilicate anion

a) Give the repeating unit of the silicate sheet.
$Si_2O_5^{2-}$

b) In a certain type of mica, one of every four silicon atoms is replaced by an aluminium atom. Suggest a formula for representing the aluminosilicate sheet.
$AlSi_3O_{10}^{5-}$

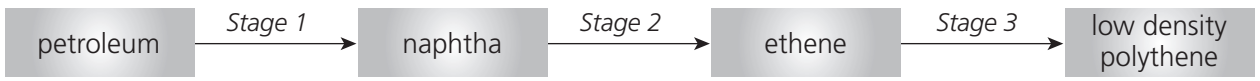
13 Kaolinite, $Al_2Si_2O_5(OH)_4$, is a 1 : 1 clay in which silicate and aluminate sheets are arranged in layers.
Refer to page T32.

a) Draw a diagram of a silicate sheet.

b) Suggest how the silicate and aluminate sheets are linked within a single layer of kaolinite.

(OCR Advanced GCE, Environmental Chemistry, Jun. 2006)

14 Low density polythene can be obtained from petroleum in three stages.
Refer to page T33.



a) Name the process for obtaining naphtha from petroleum in *Stage 1*.

b) What are the two main processes involved for obtaining ethene from naphtha in *Stage 2*?

c) Ethene forms low density polythene by addition polymerization.

i) Explain the meaning of the term 'addition polymerization'.

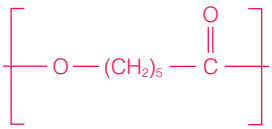
ii) The polymerization is initiated by the formation of free radicals from diacyl peroxide molecules.

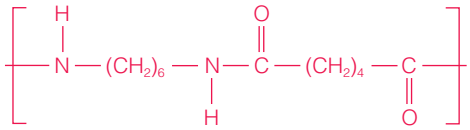
I) What is meant by the term 'free radical'?

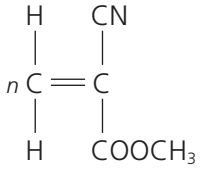
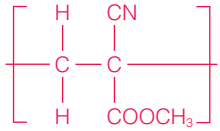
II) Outline a mechanism and give the names of mechanistic steps involved.

III) Suggest why polythene does NOT have a constant relative molecular mass.

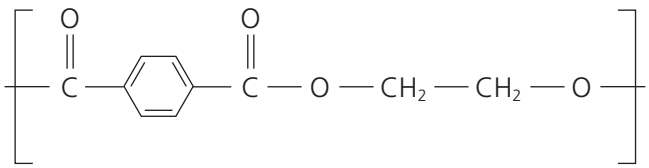
15 Write the repeating units of the following polymers.

a) $n\,HO-(CH_2)_5-\overset{\overset{O}{\|}}{C}-OH \longrightarrow$ polymer X 

b) $n\,H_2N(CH_2)_6NH_2 + n\,HOOC(CH_2)_4COOH \longrightarrow$ polymer Y 

c)  $\longrightarrow$ polymer Z 

16 Dacron is a common polyester. Its repeating unit is shown below.
Refer to page T34.



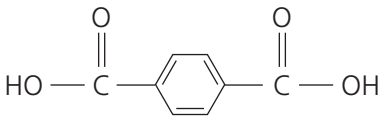
a) Suggest TWO types of material which can be made from polyesters.

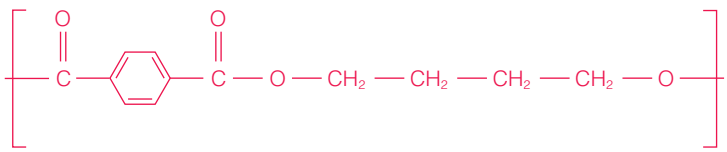
b) Draw the structural formulae of the monomers used in making Dacron.

c) Dacron is formed by a condensation reaction. Use the reaction between the two monomers drawn in (b) to explain the meaning of condensation.

d) How can Dacron be degraded in the environment?

17 Polyesters have become the major group of condensation polymers manufactured at present. There is an increasing demand for PBT, a polymer used in making mouldings for cars. PBT is made using the two monomers shown below.



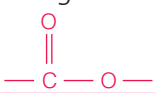
b) 



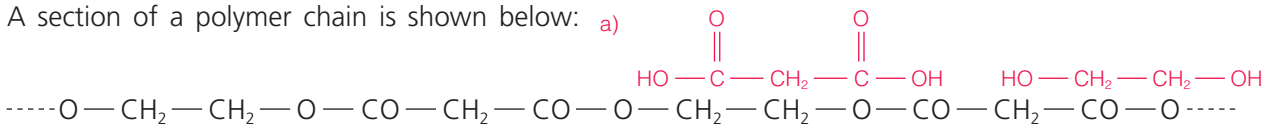
a) Explain why the formation of PBT is an example of condensation polymerization.
The monomer molecules react to form large PBT molecules, releasing water molecules.

b) Draw the repeating unit of PBT.

c) The repeating unit of PBT contains an ester linkage. Draw the structural formula of an ester linkage.



18 A section of a polymer chain is shown below:



a)

a) Give the structural formulae of the two different monomers needed to make this polymer.

b) Give the formula of the small molecule formed when the monomers combine to form the polymer.

H_2O

19 A synthetic polymer can be produced from the following two monomers:

Refer to page T34.



a) Name the type of polymerization involved in the production of the polymer.

b) Write a chemical equation for the production of the polymer.

c) Describe how a thread of the polymer can be made in the laboratory.

d) Give ONE property of the polymer and give ONE of its use which is based on this property.

e) Explain why aqueous solutions of alkalis attack the polymer readily.

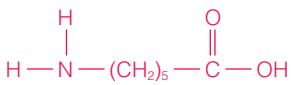
f) Explain why the disposal of waste of the polymer cause pollution problems.

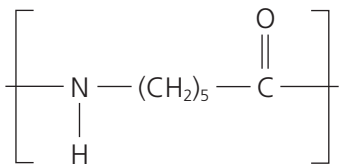
20 In the 1930s, chemists discovered nylon-6, which might replace some natural fibres.

a) Suggest a reason why chemists wanted to find synthetic replacements for some natural fibres.

b) The repeating unit of nylon-6 is shown below.

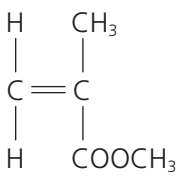
a) Any one of the following:
- Improve the properties.
- The demand is greater than nature can supply.
- Reduce the cost.

b) 

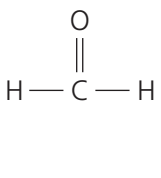


Draw the structure of a monomer which can be used to make nylon-6.

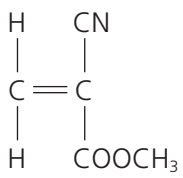
21 The structures of four compounds A, B, C and D are shown below:



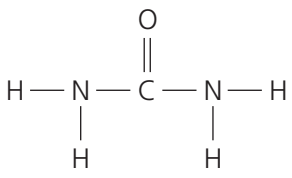
A



B



C



D

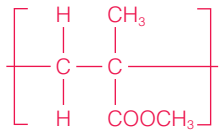
a) Under suitable conditions, compound A can polymerize to give a polymer.

i) Name the type of polymerization involved in the production of the polymer.
Addition polymerization

ii) Suggest the reaction conditions required for the production of the polymer.
In the presence of an organic peroxide at 60 °C

iii) Draw the repeating unit of the polymer.

a) iii)



iv) Is this polymer a thermoplastic or a thermosetting plastic?
A thermoplastic

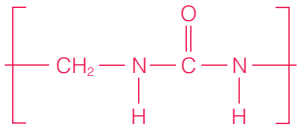
b) Compounds B and D can polymerize to give a polymer.

i) What type of polymerization the compounds undergo?
Condensation polymerization

ii) What small molecules would be eliminated?
Water molecules

b) iii)

iii) Draw the repeating unit of the polymer formed.



iv) Is this polymer a thermoplastic or a thermosetting plastic?
A thermosetting plastic

c) Compound C is the major ingredient of superglue.

i) Explain how the superglue works.
A trace of water or hydroxide ions causes an extremely rapid polymerization of the monomer, forming a rigid plastic bonded to the surfaces on which it polymerized.

ii) Name the type of polymerization involved.
Addition polymerization

22 Consider the following plastics:

polypropene, polymethyl methacrylate (PMMA), nylon, urea-methanal

Polypropene — thermoplastic
Polymethyl methacrylate — thermoplastic
Nylon — thermoplastic
Urea-methanal — thermosetting plastic

a) Classify each plastic into a thermoplastic or a thermosetting plastic.

b) Which of the plastics are produced by condensation polymerization?
Nylon and urea-methanal

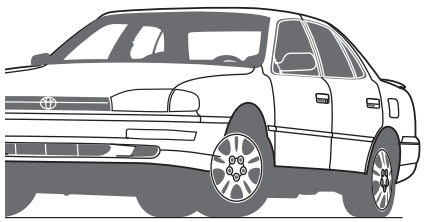
c) From the above plastics, choose one that is suitable for making each of the following items. Explain your choice in each case.



i) handle of a cooking pot



ii) stockings



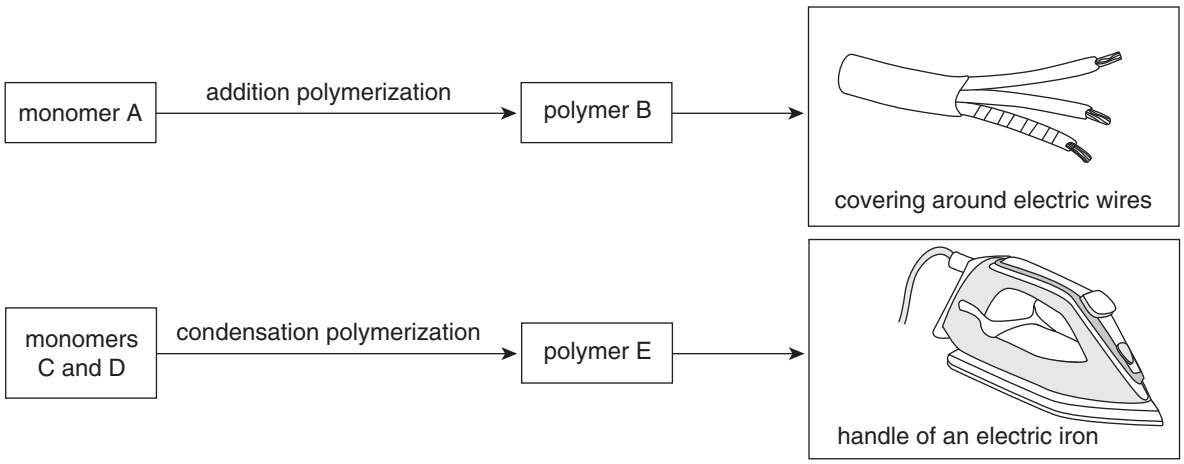
iii) lenses of car lights



iv) margarine tub

i) Urea-methanal — can withstand high temperature
ii) Nylon — strong / abrasion resistant
iii) Polymethyl methacrylate — highly transparent / strong / rigid / excellent resistance to outdoor weathering / scratch resistant
iv) Polypropene — strong / good resistance to chemical attack

23 The flow diagram below shows the production processes of two plastic articles.



a) Suggest what A and B might be.
A — chloroethene
B — polyvinyl chloride

b) Suggest what C and D might be.
C and D — urea and methanal

c) Explain the meaning of the following terms:

i) addition polymerization;

ii) condensation polymerization.

c) i) Addition polymerization is a reaction in which monomer molecules join together repeatedly to form polymer molecules.
No atoms are lost from the monomer molecules during the reaction.

ii) Condensation polymerization is reaction in which monomer molecules join together repeatedly to form polymer molecules.
Small molecules, such as H_2O, HCl or NH_3, are always formed during the reaction.

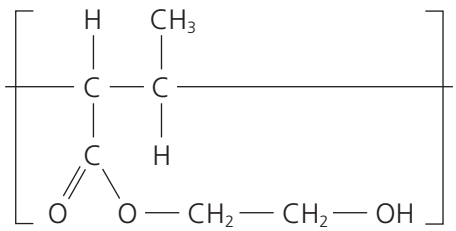
Extension

d) Suggest the reaction conditions required for the production of polymer B from monomer A.
At 40 – 80 °C in the presence of a peroxide initiator (e.g. hydrogen peroxide).

e) Suggest ONE property of polymer B that makes it suitable for making the covering around electric wires.
good insulator of electricity / flexible / easy to mould / easy to colour / waterproof / durable

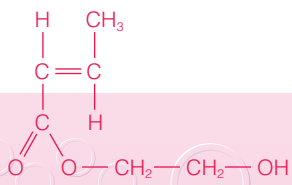
f) Suggest ONE property of polymer E that makes it suitable for making the handle of an electric iron.
Able to withstand high temperatures.

24 Contact lenses are now made from hydrogels. These polymers are able to absorb water readily. PolyHEMA is an example of a hydrogel polymer.



repeating unit in a chain of polyHEMA

a) The chains in polyHEMA are made by addition polymerization. Draw the structure of the monomer from which they are made.

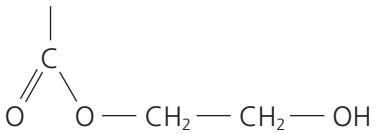


b) Suggest the type of bonding that holds the chains together.
Hydrogen bonds

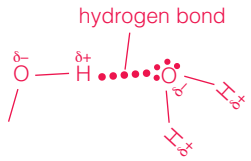
c) PolyHEMA is able to absorb water because hydrogen bonding can occur between water molecules and –OH groups on the polyHEMA structure.

Draw a diagram of a hydrogen bond formed between a water molecule and an –OH group, showing lone pairs and partial charges.

d) Alcohol groups are described as primary, secondary, and tertiary. Which name describes the alcohol group in the side-chain of polyHEMA shown below? Explain your answer.
Primary
The carbon atom bearing the –OH group has only one alkyl group attached to it.

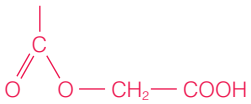


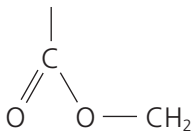
c)



e) The alcohol group in the side-chain can be oxidized to a carboxylic acid.

i) Complete the incomplete structure of the side-chain to show the carboxylic acid that is formed.





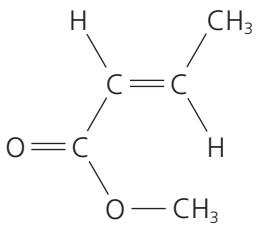
ii) Give the reactants and conditions needed to carry out this oxidation reaction in a laboratory.
Heat with acidified potassium dichromate solution under reflux.

f) Formerly, contact lenses were made from PMMA (Perspex). This polymer does not absorb water to any great extent.

Suggest why contact lenses containing absorbed water are an advantage over those that do not contain water.

f) Any one of the following:
- They are less abrasive to the eye.
- They allow gases to pass through.
- They are more flexible.
- They are more comfortable.
- They can absorb tears.
- They keep eyes moist.
- They last longer.

g) The monomer of PMMA is shown below.



PMMA monomer

i) This compound shows *cis-trans* isomerism. Describe how this kind of isomerism arises and give the name that describes the isomer shown above.
There are two different groups attached to each carbon of the carbon-carbon double bond.
Cis-trans isomers have different arrangement of their atoms in space due to the restricted rotation about a carbon-carbon double bond.

24 g) iii) Polythene is more flexible.
There are polar-polar interactions between chains of PMMA while there are only van der Waals' forces between chains of polythene.
The polymer chains of polythene can slide past each other more easily.

Extension

ii) Suggest the strongest type of intermolecular force formed between chains of PMMA.
Polar-polar interactions

iii) By comparing the intermolecular forces between the chains in PMMA and between the chains in polythene, describe and explain which of these two polymers you would expect to be the more flexible.

(OCR Advanced Subsidiary GCE (Salters), Chemistry of Natural Resources, Jan. 2005)

25 Read the passage below and answer the questions that follow.

How does non-sticky Teflon stick to cooking ware?

Teflon is a modified plastic material made from polytetrafluoroethene (PTFE). It is used as the coating of non-sticky cooking ware. The secret to Teflon's sleek (slippery) surface lies in the fluorine atoms enveloping its molecules. These fluorine atoms also repel almost all other materials, preventing them from adhering to Teflon.

'Sintering' is a technique commonly used to make Teflon stick to the surface of an item such as a cooking pan. The process of sintering is similar to melting, in which Teflon is heated to a very high temperature and then pressed firmly onto the surface of the item. One drawback of this technique is that the Teflon layer may peel off from the surface when the item is cooled to room temperature. Another technique involves chemically modifying the side of Teflon which is to be stuck onto the surface of the item. This can be done by bombarding Teflon with ions in a high vacuum under an electric field so as to break away a number of the fluorine atoms to form vacant sites, which will then be filled by oxygen-containing groups that adhere strongly to the surface of the relevant item.

a) i) Draw the repeating unit of PTFE.

ii) Is PTFE a single compound? Explain.

b) Explain, at a molecular level, why Teflon has a sleek surface.

c) Suggest why sintering can make Teflon stick onto the surface of cooking ware.

d) Suggest why Teflon with oxygen-containing groups sticks more firmly onto the surface of cooking ware.

e) Suggest ONE chemical method, not mentioned in the passage, which can make Teflon stick onto the surface of cooking ware.

f) Suggest TWO reasons for the peeling off of Teflon from cooking ware after extended use.

(HKALE 2004)